

MSc Data Analytics Dissertation Report

**Fairness Across Time: Dynamic Pricing
Framework for Edinburgh short-term rentals to
balance host revenue and student accessibility.**

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Abstract

Dynamic pricing in short-term rental markets is built to put revenue first, which means price rise sharply when demand is high which squeeze out budget conscious travelers. My main purpose is to study Edinburgh's Airbnb market over June, July, and August. I picked this specific time framework because this is the realistic window to travel for working students all over UK. Using publicly available from Inside Airbnb data i.e. 8,026 entire-home unique listings, 730,461 listings/nights across summer, our study develops a fairness aware dynamic pricing framework. We use Random Forest regression model and predicts nightly prices from listings and temporal features (test $R^2 = 0.39$, RMSE = £76, MAE = £47). We get predictions prices that we feed into a simulation by applying a tunable fairness parameter (λ) to balance host revenue against student accessibility.

Our Descriptive analysis shows that only 22.7% of summer listings/nights are affordable at £75/night student budget threshold. Affordability falls to 14% in August and 5.8% in central Low access neighborhoods' during August. We also did Kruskal-Wallis statistical test that confirms prices differ significantly across accessible categories and summer months when we got the same value $p < 0.0001$. The fairness simulation uses a prioritized compliance mechanism. At each value of λ , a specific share of listings (price above £75) is reduced to comply to our threshold. The resulting trade-off frontier is monotonically concave on the:

- Revenue change
- Affordability gain

The concavity is verified by an automated check in simulation code.

Recommended operating points:

- $\lambda \sim 0.45$, within sweet-spot band of 0.4 – 0.5 which is shown in figure: Trade-off figure, Affordability rises from 22.7% to about 28% while host revenue falls by less than 1.1%.
- $\lambda = 1.0$,
Affordability lifts up to 35.2% while host revenue only falls by 7.8%.
This is much larger price for a small extra gain.

Monthly disaggregation shows that August requires roughly twice the host-revenue adjustment of June to achieve comparable affordability gains, supporting a stronger intervention during the festival period. The findings remain consistent under two independent baselines (observed median prices and Random Forest predictions) and an XGBoost cross-check ($R^2 = 0.45$). Trade-off curves are robust across three student-budget thresholds (£60, £75, £90). The framework's main contribution is to adapt fairness constraints from classification to a continuous-pricing context, producing a month-sensitive, policy-relevant simulation that balances host revenue against student affordability in Edinburgh's short-term rental market. The framework is presented as a proof-of-concept policy simulation rather than evidence of real-world intervention effectiveness.

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Headline Findings

1. **Affordability gap** is real and measurable.
Across Edinburgh's short-term rentals during summer, only 22.7% of listings/night fall within £75/night student budget.
2. **Problem worsens sharply during festival season.** Affordability collapses to 14% in August leaving a particular budget group out when the city is most active.
3. **Modest pricing intervention can substantially close the gap.** When framework's fairness weight (Λ) is set to recommended value of 0.45 which uplifts affordability by 5-6% when host revenue falls by less than 1.1%.
Trade-off is favorable: small revenue discount produces meaningful affordability gain.
4. **Results are robust, not artifacts of method choice.** Same conclusion emerges under:
 - two independent pricing baselines (stratified observed median and Random forest prediction whose curves overlap)
 - XGBoost cross-check
 - three budget thresholds (£60, £75, £90)Indicates that framework's findings are stable rather than sensitive to specification.

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1. Executive Summary

Area	Status	Deliverables
1. Introduction	Complete	Research questions, aim, objectives defined.
2. Literature Review	Complete	25+ papers collected, fully reviewed and analysed.
3. Methodology	Complete	Two-baseline simulation; prioritized compliance mechanism; XGBoost robustness check.
4. Data Collection	Complete	Inside Airbnb listings.csv and calendar.csv for Edinburgh obtained.
5. Data Cleaning	Complete	8,026 entire-home listings retained; £ 30 – 500 hard cutoff applied.
6. Neighbourhood classification	Complete	4 category student accessibility mapping (high, moderate, low, outer).
7. Student budget threshold	Complete	Central value £ 75/night; sensitivity tested at £ 60 and £ 90.
8. Descriptive Analysis	Complete	22.7% of summer nights affordable at £ 75; group differences statistically significant.
9. Random Forest model	Complete	$R^2 = 0.39$, $RMSE \approx £76$ across 627,810 modelling rows.
10. XGBoost robustness	Complete	$R^2 = 0.45$, $RMSE = £72$; same fairness trade-off curve as RF.
11. Fairness simulation	Complete	λ sweep from 0 to 1 across two baselines; monotone concavity verified.
12. Sensitivity Analysis	Complete	Tested across budget thresholds (£60/£75/£90) and cap multipliers (1.2×, 1.3×, 1.5×).
13. Policy Dashboard	Complete	Single-page executive view; recommended $\lambda = 0.45$.
14. Ethics	Complete	Low-risk secondary-data screening accepted.
15. Project Management	Complete	Timeline and risk register updated.

2. Project Overview

2.1 Introduction and Project context

The rapid growth of short-term rental platforms, especially Airbnb, has transformed urban housing markets around the world. (Wachsmuth & Weisler, 2018) worked with New York city data. This demonstrate that short-term rental platforms introduce new revenue flow into housing markets which is geographically uneven and concentrated in culturally desirable neighbourhoods. (Gago & Cocola-Gant, 2021) reach a complementary conclusion in Lisbon which shows that platforms accelerate neighbourhood transformation by giving landlords a financial route to pull properties out of long-term market.

Edinburgh is one sharp example of these pressures. Capital city of Scotland, UNESCO World heritage site, this city draws around 4 million tourists every year. Demand peaks during August Festival Fringe and Hogmanay (Scottish-Government, 2019). At the same time, Edinburgh hosts more than 50,000 students across different universities inside the city. Those students need affordable housing in close proximity of campus which includes: Marchmont, Bruntsfield, Newington, Southside to name a few. Foundational UK academic work cited by Scottish government, (Rae, 2018), has documented that Edinburgh's city center ward alone accounted for 8.5% of Scotland's active Airbnb listings, and (Crawford, et al., 2023) confirm that festival-driven demand concentrates precisely in the neighbourhoods where student housing pressure is greatest.

This study focuses on a specific and underexplored dimension of our problem: whether Edinburgh's short-term rental market remains accessible during the summer months for students visiting from other UK cities. Students resident in cities such as Manchester, Leeds, Birmingham, and Nottingham, all of which contain large student populations, represent a group with constrained budgets, high sensitivity to accommodation costs, and a clear rationale for summer travel to Edinburgh. As maintenance loans and casual earnings impose a genuine ceiling on nightly expenditure, revenue-maximising dynamic pricing algorithms systematically disadvantage this group precisely when affordability gap is widest. This study proposes a fairness-aware framework that makes this trade-off explicit and quantifies the conditions under which it can be moderated.

Visiting students are selected as target group for several reasons. First, their budgets are structurally constrained and are therefore particularly sensitive to accommodation cost shocks during peak-demand periods. Second, summer mobility for educational purposes, conference attendance, and cultural travel is a recognised dimension of student experience that existing policy frameworks have largely overlooked. Third, spatial concentration of Edinburgh's festival demand in student-adjacent neighbourhoods (Marchmont, Bruntsfield, Newington, and Southside) means that visiting students and festival tourists compete directly for same limited rental supply, making this group a meaningful and tractable target for fairness intervention.

2.2 Problem Statement

The research problem had three dimensions:

1. Dynamic pricing research is overwhelmingly geared toward revenue optimization. (Xie & Chen, 2017) and (Wang & Nicolau, 2017) both offer detailed analyses of pricing behavior in sharing economy but their models are built entirely around maximizing host income. They do not feature equity and accessibility for particular groups. (Gibbs, et al., 2018) examined dynamic pricing adoption across five markets and observed that algorithmic systems allow hosts to respond quickly to demand shocks. Social welfare plays no part in these frameworks.
2. Literature on fairness in machine learning has primarily concentrated on classification issues: recruitment, financing, criminal justice (Mehrabi, et al., 2021) (Dwork, et al., 2012) (Hardt, et al., 2016) rather than on pricing algorithms. There has been no effort to apply fairness constraints to seasonal pricing. This is significant because **pricing and classification** differ structurally in ways that influence fairness in application. Prices are continuous as opposed to binary and **they** affect both buyers and sellers. Both pricing and classification sit inside markets that will shift in response to any constraint imposed.
3. Although there are studies on short-term rental in Edinburgh (Rae, 2018) , (Crawford, et al., 2023), no research has addressed accessibility for student visiting from other cities in UK during high demand season.

2.3 Research Questions and Hypotheses

RQ1: How much more do summer Airbnb prices in Edinburgh cost in compare to what students can afford and how does this affordability difference change across different areas and months?

Null Hypothesis (H01)	Alternative Hypothesis (H11)
There is no difference between summer Airbnb prices in areas where students live and £75 / night budget for students and it does not change much across areas or months.	Summer Airbnb prices in areas where students live are much higher than £75 / night budget with difference in August and in areas close to universities like: Marchmont, Bruntsfield, Newington and Southside. We also tested prices at £60 and £90 to make sure our results are correct.

RQ2: What is the trade-off between host revenue and student accessibility when summer pricing is subject to fairness limits?

Null Hypothesis (H02)	Alternative Hypothesis (H12)
There is no quantifiable trade-off between host income maintenance and improved student accessibility when applying fairness penalties (λ) to the pricing.	Revenue maintenance and accessibility development have a concave trade-off frontier when increasing fairness penalties beyond a certain point (around $\lambda \approx 0.4$ – 0.5) yields diminishing returns.

RQ3: What is the ideal fairness penalty parameter (λ) that balances revenue maximisation and student access during Edinburgh's summer?

Null Hypothesis (H03)	Alternative Hypothesis (H13)
The ideal λ value is same for all months and neighbourhoods during summer months.	The optimal λ (within the tested range 0–1) varies by month and neighbourhood. August requires stronger fairness interventions (a higher λ) than June or July owing to greater demand pressure. The optimisation is scoped as a proof-of-concept demonstration rather than a full-scale optimisation system.

2.4 Research Aim and Objectives

2.4.1 Aim

The goal of this research is to design and evaluate a pricing system for short-term rentals in Edinburgh. This system aims to be fair and find a balance between helping hosts earn money and making rentals affordable for students who visit during summer.

2.4.2 Objectives

No.	Description	Linked RQ
1.	I want to look at the prices of Airbnb places in Edinburgh during the summer. I will make a map of how much they cost in areas of the city. This will help me see what people usually pay in June, July and August.	RQ1
2.	I will compare the summer prices to a price of £75 per night. This is what I think is affordable for students after evaluating from different sources. I will also look at what happens if the price is £60 or £90 per night.	RQ1
3.	I will use a kind of computer program to predict how much Airbnb places will cost. This program is called a Random Forest model. I will test it to see how well it works.	RQ2
4.	I will check my results with another model XGBoost. This will help me make sure my results are correct.	RQ2

5. I will make a system that tries to be fair to everyone. This system will have a setting that I can adjust. I will test it with two groups of data. RQ2, RQ3
6. I will look at how much money people can save and how much more affordable the prices are. I want to find the point where it does not make sense to adjust the prices anymore. RQ2
7. I will look at results for each month. I will also try prices and see what happens. RQ3
8. I will use what I have learned to make suggestions for city of Edinburgh. I want to help make decisions about Airbnb and tourism All RQs

2.5 Scope and Delimitation

Dimension	Scope	Justification
Geographic	Edinburgh council area	We have a lot of data for the Edinburgh council area. It is a clear example to study.
Platform	Airbnb only	Airbnb is a leader in the short-term rental market and it is easy to get data from them.
Time period	Summer months (June, July, August)	These are realistic time period for student to travel and prices are really high.
Student cities	Manchester, Leeds, Nottingham, Birmingham	These cities have a major student population and have similar urban context.
Fairness simulation	Students vs. tourists (neighbourhood proxy)	We are looking at how price can be fair for students compared to tourists and we are using the neighborhood as a way to figure this out. This is something we can measure. It is important for making policies.
Methodology	Quantitative simulation	We are using numbers to simulate things and see what happens. We want to see if one thing causes another to happen.
Baseline	Stratified observed medians + RF predictions (XGBoost cross-check)	We are using Random forest model to make predictions and we are double checking with XGBoost. We want to make sure our baseline is strong and accurate.

What this study does not claim to be doing:

- » This study does not include travel websites like Booking.com or Expedia. The reason is that their data is not available.
- » It also does not involve testing on platforms.
- » The study only looks at students who visit during the summer.
- » The findings may not apply to times of the year.
- » It also does not consider how the housing market changes, over time.
- » The study focuses on students who are visiting not those who live there.
- » It does not consider how hosts might change their pricing strategies.

2.6 Stakeholder Analysis

The fairness framework affects different groups of people, and these groups don't all want the same thing. Recognising this diversity shows that the study isn't just a technical exercise; it actually has something useful to say about real-world policy decisions.

Stakeholder	Primary objective	Impact of a fairness constraint (λ)	Position on intervention
Hosts (professional)	Maximise nightly revenue across portfolio; use dynamic-pricing tools.	Modest revenue loss at recommended $\lambda \approx 0.45$ (–1.1%). Larger loss only at extreme λ .	Likely opposed unless platform mandates uniform compliance; otherwise distorts competition.
Hosts (casual)	Supplementary income; static or rule-based pricing.	Negligible direct impact; they rarely operate at the aggressive end of the price distribution.	Largely neutral.
Platform operator (Airbnb)	Maximise bookings $\times$ take rate; protect market share.	Higher occupancy via more affordable inventory may offset host revenue loss; reputational gain on accessibility.	Potentially supportive if integrated as a user-selectable mode (see section 6.7).
Edinburgh City Council	Manage housing affordability, festival economy, residents' quality of life.	Pricing-based lever complements existing 90-day rules and licensing schemes.	Strongly supportive, provides a tunable policy tool short of bans.
Local residents	Stable rents; reduced displacement during peak season.	Indirect: lower short-term rates may reduce landlord incentive to switch from long-term to short-term lets.	Supportive in principle; effect is second-order.
Visiting students	Affordable summer accommodation in Edinburgh.	Direct beneficiary's affordability rises from 22.7% to ~28% at recommended λ .	Primary beneficiary.
Universities (UK)	Student mobility, summer programs participation.	Indirect: improved access to host city for visiting cohorts, summer schools, conferences.	Mildly supportive.

There are two things that stand out from this matrix.

1. First company that runs the platform has a position than people who host. The company can handle a change in how money is distributed if it means more people use the platform and it gets a good reputation.
2. The professional hosts are the group that really does not want this to happen. We can make them follow the rules by making sure everyone has to comply than just leaving it up to them. If we do not do this the hosts who do not follow the rules will just take business away, from the ones that do.

These considerations inform implementation pathway discussed in Section 6.7.

3. Literature Review

This chapter sets out theoretical and empirical foundations of my study. This has nine sections:

- 3.1) unifying theoretical framework that links five conceptual ideas applied throughout.
- 3.2) impact of short-term rental platforms on urban housing.
- 3.3) dynamic pricing in sharing economy
- 3.4) student housing accessibility
- 3.5) fairness in algorithmic systems
- 3.6) regulatory approaches to short-term rentals across comparable cities
- 3.7) Edinburgh context with particular attention to festival economics
- 3.8) research gap identification
- 3.9) conceptual framework that operationalize these strands into study's methodology

3.1 Theoretical Framework: platform pricing, inequality, governance

Five conceptual lenses inform this study and reinforce each other on the question of how digital platforms reshape access to scarce urban resources. **Platform capitalism** (Srnicek, 2017) describes the economic model in which platforms extract value by intermediating transactions between independent parties; Airbnb functions not as a passive marketplace but as an active price-setter through its dynamic pricing and ranking algorithms. The λ parameter introduced in this study can be read as a deliberate intervention into this model, encoding a distributional objective the market does not generate on its own.

Digital inequality (Edelman, 2015) treats platform pricing as a site where pre-existing income asymmetries are amplified rather than neutralised: revenue-optimising algorithms systematically disadvantage groups with constrained budgets, including students whose budgets are bounded by maintenance loans and casual earnings. **Algorithmic governance** (Yeung, 2018) frames pricing algorithms as private regulatory instruments; existing public regulation (Edinburgh's 90-day rule, Barcelona's licensing scheme) operates on the quantity side, while the λ -framework introduces a complementary price-side lever.

Urban tourism economics treats festival-driven demand as a textbook demand shock against fixed short-term supply: Edinburgh's four-week festival window concentrates global cultural demand in a supply-constrained city, and (Gibbs, et al., 2018) document that hosts in high-demand leisure markets adjust prices most aggressively. Finally, housing commodification (Aalbers & M.B., 2016); (Wachsmuth & Weisler, 2018) describes the treatment of dwellings as financial assets rather than use-values, with short-term rentals accelerating the financialisation of residential space. Edinburgh's traditional student neighbourhoods sit at the overlap of commodification pressure and student housing demand.

Read together, these five lenses produce a coherent positioning: the present study introduces a continuous, tunable fairness mechanism into a system (algorithmic platform pricing) that has so far operated under a single objective (host revenue) reinforced by a particular economic logic (peak-demand commodification). The contribution is both conceptual, adapting fairness from classification to continuous-output contexts, and methodological, in making the distributional parameter (λ) explicit and operationally controllable within a reproducible simulation.

3.2 Short-term rental platforms and urban housing impacts

(Guttentag, 2015) provides foundational account of Airbnb as disruptive innovation in Christensen's sense, one that competed initially at lower end of the accommodation market before gaining mainstream traction through payment infrastructure, host professionalisation, and trust mechanisms. This trajectory is relevant because it explains why affordability pressure is not marginal but structural: platform's pricing behaviour now drives, rather than follows, urban accommodation markets.

Several studies converge on the conclusion that Airbnb growth has distributional consequences for urban housing (Edelman, 2015) demonstrate through a field experiment that platform design choices can entrench discrimination — guests with distinctively African-American names were 16% less likely to be accepted than equivalent guests with distinctively white names, establishing that platforms encode rather than neutralise offline biases. (Wachsmuth & Weisler, 2018) develop a rent-gap framework for New York City, arguing that short-term rentals introduce a new revenue flow into housing markets that is systematic but geographically uneven, creating new pressures in culturally desirable neighbourhoods. (Gago & Cocola-Gant, 2021) a parallel conclusion through qualitative fieldwork in Lisbon, framing Airbnb as buy-to-let gentrification experienced locally as social injustice. Crucially, none of these studies moves beyond diagnosis to propose pricing-based remedies, the gap this dissertation seeks to address.

Within UK specifically, (Rae, 2018) and the (Scottish-Government, 2019) research into the impact of short-term lets on Scottish communities together provide the empirical foundation for the Edinburgh case. Rae documents how growth of short-term lets a highly localised phenomenon: a small group of "ultra-desirable" localities sit at the epicentre of a newly globalised market. With Edinburgh's City Centre ward alone accounting for 8.5% of all active Airbnb listings in Scotland by May 2019. These papers validate both data source and spatial framing of the present analysis.

3.3 Dynamic pricing in sharing economy

Research on dynamic pricing in the sharing economy splits into two strands: descriptive studies examining how prices are determined, and prescriptive models aimed at maximising revenue. Neither has substantively engaged with the question of who is priced out as a result (Xie & Chen, 2017) decompose price variation into property characteristics, host attributes, and location factors via a hedonic approach, accounting for approximately 60% of the variance. (Wang & Nicolau, 2017) extend this to 180,533 listings across 33 cities and identify twenty-five explanatory variables, establishing that cross-city generalisation requires caution. (Gibbs, et al., 2018) provide the most comprehensive analysis of dynamic pricing adoption, demonstrating that uptake is strongly heterogeneous: professional multi-listing operators in high-demand leisure markets vary prices most aggressively. This professional-versus-casual divide is directly relevant to fairness intervention design, since a uniform constraint will burden these groups differently.

On the prescriptive side, (Breiman, 2001) formalised Random Forests and (Chen & Guestrin, 2016) introduced XGBoost; both methods are now widely deployed in pricing pipelines. In the

present study, these models serve a specific and bounded role: they provide a model-based pricing baseline that complements the stratified observed-median baseline. The near-identical trade-off frontiers produced by both baselines confirm that the fairness conclusions are not contingent on the choice of prediction model. The models are not the primary contribution of the study; rather, they function as robustness instruments for the fairness simulation. Both methods are conventionally evaluated through RMSE, MAE, and R^2 ; metrics that say nothing about distributional outcomes. (Lundberg & Lee, 2017) proposed SHAP as a unified framework for model interpretability, but interpretability is not equivalent to fairness: revealing what a model does does-not constrain it toward equitable outcomes. The gap this study addresses is the absence of a mechanism that explicitly encodes and navigates the revenue–accessibility trade-off.

3.4 Student housing accessibility

The phrase "studentification" was first used by (Hubbard, 2009) to characterise how student concentrations alter area housing markets in four ways: economically, culturally, physically, and demographically. His UK case studies, which concentrated on Loughborough but theorised for the larger sector, demonstrated that neighbourhoods with high student populations typically face localised rent inflation and the conversion of family homes into shared apartments; these circumstances make these neighbourhoods particularly vulnerable when short-term rentals enter the picture. Hubbard's observation that seasonal rhythms of student life create summer vacancies that landlords can exploit has direct relevance for this study's focus on summer Airbnb supply in student-adjacent neighbourhoods.

More recent work has documented the deepening of student housing pressures in the UK. As (Scottish-Government, 2019) research records, Edinburgh's concentrated student population overlaps spatially with the wards most affected by short-term-let growth. It is a structural pattern that creates direct competition between long-term student tenants and short-term festival guests during peak months. The present study focuses on a group distinct from residential students: visiting students who travel from other UK cities over summer and compete directly with festival attendees for short-term accommodation. This population falls outside the protection of residential tenancy safeguards and is underserved by existing studentification literature, which has concentrated on neighbourhood effects for permanently resident students. Addressing this gap, and demonstrating the scale of affordability barrier these visitors face, is one of our study's contributions.

3.5 Fairness in algorithmic systems

The literature about fairness in machine learning talks about two ideas. One is fairness, which is about treating similar cases in a similar way. The other is group fairness, which's about making sure that different groups of people are treated equally. For pricing context, these distinctions translate into questions about whether similar listings are priced consistently and whether specific demand groups face systematically worse access than others. Some researchers, like (Chouldechova, 2017) and (Kleinberg, et al., 2017) have shown that it is not possible to make everyone equal in every way especially when different groups have starting points, reinforcing the case for making trade-offs explicit rather than claiming a single correct fairness outcome. This study adopts this perspective directly: rather than asserting one fairness criterion, it employs the λ parameter to navigate the revenue–accessibility frontier transparently.

3.5.1 Comparing fairness criteria for pricing context

The algorithmic-fairness literature frequently uses four fairness criteria, all of which were created mainly for classification. Each criterion, its mathematical foundation, and whether or not it translates into a continuous-pricing scenario are summarised in the table below.

Fairness criterion	Core definition (classification context)	What it would require in pricing	Suitability for this study
Demographic parity (Dwork, et al., 2012)	Equal positive-decision rates across groups regardless of qualifications.	Equal share of affordable listings across all demand groups, irrespective of property attributes or neighbourhood.	Too rigid: ignores legitimate price drivers (size, location, season). Would over-correct or be politically infeasible.
Equality of opportunity (Hardt, et al., 2016)	Equal true-positive rates across groups; same access for those who "qualify".	Equal access (affordability rate) for the user group whose budget falls within a defined window.	Conceptually closest to this study's framing: students with budgets in the £60–90 band should not face systematically worse availability than other guests with the same budget. But the binary classification machinery does not translate cleanly to continuous prices.

Welfare optimization (Barocas, et al., 2019) (Mehrabi, et al., 2021)	Maximise aggregate utility subject to a fairness constraint; tunable weighting between groups.	Maximise host revenue subject to a minimum affordability floor for the target group.	Maps onto the simulation: every λ defines an attainable (revenue, affordability) point, and the trade-off frontier traces the welfare surface.
Proportional fairness (Chouldechova, 2017) (Kleinberg, et al., 2017)	Balanced trade-offs in the face of impossibility results: no party can be made better off without proportionally compensating the affected group.	A tunable balance where each unit of intervention is justified by a proportional gain to the disadvantaged group.	This is precisely what the λ parameter operationalises: λ encodes how much revenue loss is acceptable per unit of affordability gain, with the sweet-spot band defined by where the marginal trade-off remains favourable.

3.5.2 Why classification fairness does not transfer directly

Where the literature falls short, however, is in its almost exclusive focus on classification problems: hiring, lending, recidivism scoring. Pricing is a fundamentally different setting. Outcomes are continuous rather than binary, both buyers and sellers are affected, markets adjust dynamically to any constraint imposed, and feedback loops between price and demand are immediate. (Dwork, et al., 2012) developed Lipschitz conditions for individual fairness, (Hardt, et al., 2016) introduced threshold-based post-processing for equality of opportunity. Both were designed for discrete decision boundaries, and neither transfers straightforwardly to a continuous pricing variable. (Barocas, et al., 2019) acknowledge in their standard reference that fairness definitions are inherently normative, encoding different values but their practical guidance remains centred on classification. The question of what fairness means when output is a nightly rate rather than an accept/reject decision remains largely unaddressed, which is precisely where this study contributes.

3.6 Regulatory approaches to short-term rentals across cities

Over the past decade, short-term rental regulation has emerged as a distinct policy domain, with cities deploying diverse instruments. (Aguilera, et al., 2021) compare regulatory responses in Barcelona, Paris, and Milan, identifying three regulatory intensities: weak (Milan), intermediate (Paris, with a 120-day primary-residence rule and per-rental tax), and strong (Barcelona, with strict zonal licensing). (Bei & F., 2023) expand this comparative framework using a counterfactual assessment of European cities, finding that stricter market-access restrictions (registration, licensing, zonal caps) produce measurable reductions in listings and in the share of professional multi-listing hosts. Data-sharing agreements between platforms and regulators, active in Amsterdam, Barcelona, London, and Paris, assist enforcement without independently altering supply dynamics.

City	Headline regulatory instrument	Mechanism	What it does not address
Barcelona	Strict zonal licensing tied to housing pressure; planned 2028 phase-out of tourist STRs.	Caps STR supply in central zones; no new licences in saturated areas.	Does not act on prices charged by the listings that remain compliant.
Amsterdam	30-night annual cap; mandatory registration; in some areas, outright bans.	Caps how often a property can be let on a short-term basis.	Caps quantity but not price; affluent guests still outbid budget-constrained ones.
London	90-day rule for entire-home short lets in Greater London; new UK-wide STL register from 2024.	Limits annual short-let nights per property; introduces transparency.	No price-level intervention; festival cities outside London (e.g. Edinburgh) have additional regimes.
New York City	Local Law 18 (2023): mandatory registration, host presence, two-guest cap.	Effectively eliminates entire-home short lets; reduced listings by ~80% by mid-2024.	Severe access reduction without a graduated mechanism for moderate intervention.
Edinburgh	Short-Term Let Control Area (city-wide); mandatory licensing; 5% visitor levy from 2026.	Combination of quantity caps and tax instruments.	No price-level lever to address festival-period affordability for specific user groups.

Reading across these cases, a clear pattern emerges: every regulatory regime currently in place operates on the supply side, restricting how many short-term rentals can exist or how often they can be used, rather than on the price they charge. This makes the present study's contribution structurally distinct. The λ -framework proposes a pricing-based lever that complements rather than replaces existing toolkit, addressing the affordability dimension that quantity-only regulation leaves untouched.

3.7 Edinburgh Context

Edinburgh offers a particularly revealing case for studying short-term rental dynamics, given its unique combination of a globally significant festival season, concentrated student populations, and a compact urban geography where tourist and residential areas overlap.

(Crawford, et al., 2023) provide most directly relevant academic analysis of Edinburgh's short-term-let sector. Using open Inside Airbnb data, same data source used by this dissertation to construct a neighbourhood-level vulnerability index. Their index, built from comparisons between estimated short-term-let income and long-term rental market values, identifies neighbourhoods at greatest risk of property switching from long-term to short-term use, and confirms that spatial pattern of vulnerability concentrates in central and student-adjacent areas this dissertation designates as priority zones for fairness intervention.

(Rae, 2018) provides foundational geographic analysis of short-term lets in Scotland, characterising the rise of Airbnb as a "global hood" phenomenon: a small group of ultra-desirable localities forming the epicenter of a globalised short-term-rental market. (Scottish-Government, 2019) research into the impact of short-term lets on Scottish communities operationalises Rae's framework with empirical detail, showing that just over half of all active Airbnb listings in Scotland in May 2019 were in the City of Edinburgh or Highland council areas, that the City Centre ward alone hosted 2,710 active listings (8.5% of the Scottish total). The same research notes the spatial concentration in Edinburgh's Old Town, contained 812 active Airbnb listings, among the highest densities recorded anywhere in the dataset.

The festival hosts in this area form a distinct professional cohort specialising in peak-period lets at materially higher price points than non-festival operators. Edinburgh's traditional student areas: Marchmont, Bruntsfield, Newington, and the Southside, sit adjacent to this festival epicentre, combining walkable proximity to campus with the period architecture that attracts visitors. This geographic overlap creates direct competition between student visitors and festival tourists for the same limited rental supply, which fairness framework directly addresses.

3.8 Research gap identification

Three related gaps are identifiable from the literature.

- » First, dynamic pricing research has not engaged with equity or affordability for specific demand groups; all major models optimise exclusively for host or platform revenue.
- » Second, fairness in machine learning has been applied almost entirely to classification contexts and does not yet provide workable frameworks for continuous-pricing settings.
- » Third, while Edinburgh's short-term-let market has been studied at neighbourhood level (Crawford, et al., 2023); (Rae, 2018); (Scottish-Government, 2019), no prior work has quantified affordability barrier facing visiting students or evaluated revenue cost of price-based fairness interventions.

The present study addresses all three gaps by adapting group fairness principles from classification to a continuous-pricing context, applying them to Edinburgh's summer rental market, and producing a policy-ready simulation framework with quantified trade-off outcomes.

3.9 Conceptual Framework

3.9.1 Components

The conceptual framework integrates three analytical layers:

- (i) a descriptive layer that characterises affordability gap across neighbourhoods and months;
- (ii) a modelling layer that generates baseline price predictions using Random Forest and XGBoost regression;
- (iii) a simulation layer that applies the prioritised compliance mechanism across a range of λ values to produce trade-off frontier. Each layer feeds the next, and the outputs are integrated into a policy dashboard designed for stakeholder engagement.

3.9.2 Mathematical Formulation

The framework is operationalised through two adjustment mechanisms — a headline mechanism and a sensitivity alternative, that share the same conceptual structure but differ in how the fairness constraint binds.

Prioritised-compliance mechanism (headline).

Let $B = £75$ denote the central budget threshold, and let U denote the set of accessible-neighbourhood listing-nights priced above B (i.e. nights for which $\text{price} > B$ and $\text{access_cat} \in$

{High, Moderate}). At $\lambda \in [0, 1]$, a share λ of the rows in U is required to comply, with their adjusted price set to B . The remaining $(1 - \lambda)$ share retains its original price. Compliance is allocated by sorting U so that the highest-priced listings comply first, reflecting a policy preference for targeting the most aggressive overpricing.

Formally, for a row $i \in U$ with rank $r(i) \in [0, 1)$ within U (highest price = rank 0):

$p_{\text{adj}}(i) = B$ if $r(i) < \lambda$; $p_{\text{adj}}(i) = p(i)$ otherwise.

For rows outside U (already affordable, or outside accessible neighbourhoods), $p_{\text{adj}} = p$.

Stratum-anchored mechanism (sensitivity).

The earlier draft used a continuous linear-blend mechanism that adjusted prices toward a neighbourhood-month cap rather than the budget threshold:

$p_{\text{adj}} = p - \lambda \times \max(0, p - p_{\text{fair}})$,

where $p_{\text{fair}} = \min(p, m \times \text{median}_{\{\text{category}, \text{month}\}})$ and $m \in \{1.2, 1.3, 1.5\}$. Under the prioritised-compliance design, this mechanism is reported as sensitivity only because pricing toward the stratum cap (approximately 1.3× the local median, typically £125–£250 in August) never reaches the £75 student budget at any λ , producing zero affordability gain in the simulation.

Interpretation of λ .

At $\lambda = 0$ the model behaves as a pure revenue maximiser. At $\lambda = 1$, fairness carries its full weight and every over-budget listing in accessible neighbourhoods is required to comply. Intermediate values map onto partial intervention positions where only the most extreme overpricing is targeted first. The simulation tests λ at 0.1 increments (eleven points), producing a clean trade-off frontier per baseline and per sensitivity slice.

4. Methodology

4.1 Research philosophy and approach

The study sits within a positivist tradition. The core assumption is that pricing behaviour, demand patterns, and affordability outcomes in Edinburgh's short-term rental market can be quantified and modelled from observable data. The approach is deductive: hypotheses drawn from the literature on dynamic pricing and housing affordability are tested through empirical analysis and simulation. The design is fully quantitative, relying on publicly available secondary data. The decision to use simulation rather than real-world implementation reflects both the project timeframe and the practical reality that no platform operator would trial an untested pricing intervention on the basis of a student project.

4.2 Data sources and collection

Two datasets form the backbone of our analysis:

Dataset	Source	Description	Key variables	Expected size
listings.csv	Inside Airbnb (via Kaggle)	Property-level details for Edinburgh listings.	price, room type, neighbourhood, bedrooms, accommodates, host info	~13,246 listings
calendar.csv	Inside Airbnb (via Kaggle)	Availability and price data for each listing across dates.	date, listing_id, price, adjusted_price, available	~4.8 million rows

Data are obtained from Inside Airbnb, a publicly available repository that scrapes Airbnb listing and availability data for major cities at regular intervals. Pricing are recorded for each listing over a twelve-month window (June 2019 to June 2020). The analysis is restricted to summer period (June, July, and August 2019), which predates the COVID-19 disruption and represents a structurally normal festival season. No primary data collection was conducted; the data are publicly available and already visible on the Airbnb platform, raising no personal data concerns.

4.3 Data Cleaning and Preprocessing

Each step is documented with details on how many rows were removed and why.

Step	Rule Applied	Rationale	Records After
Price parsing	Strip currency symbols and commas; convert to float.	Raw prices stored as strings in Airbnb format.	—
Room type filter	Retain only "Entire home/apt".	Visiting students would book a whole unit rather than a private or shared room; consistent with the accessibility framing.	—
Price hard cutoff	Exclude listings with nightly price < £30 or > £500.	Eliminates data-entry errors and extreme outliers while retaining all economically plausible prices; a standard cleaning step in the Edinburgh STR literature (Crawford et al., 2023).	8,026 unique listings
Missing values	Drop rows missing price or neighbourhood; impute bedrooms from accommodates field where feasible.	Preserves the maximum usable sample while maintaining analytic integrity.	—
Neighbourhood mapping	Map all Edinburgh neighbourhoods to four accessibility categories (High, Moderate, Low, Outer) by exact string match.	Provides an operationalisable proxy for student proximity to campus. Unmapped neighbourhoods default to Outer.	4 categories
Calendar merge	Inner join calendar to listings on listing_id = id; filter to June–August 2019.	Produces the summer panel used in all descriptive and simulation analyses.	730,461 listing-nights

Category	Neighbourhood values	Justification
High student accessibility	Meadows and Southside; Marchmont and Sciennes; Newington and Dalkeith Road; Bruntsfield and Morningside (southern portion)	Traditional student areas clustered around the University of Edinburgh, where student housing density is highest.
Moderate accessibility	South Leith; Tollcross; Fountainbridge and Craiglockhart; Leith Walk and Pilrig	Mixed areas with some student residents; reasonably close to campuses but not dominated by student housing.
Low accessibility	Old Town, Princes Street and Leith Street; New Town West; New Town East; Canongate, Southside and Dumbiedykes; Dean and Stockbridge	Central tourist zones with premium pricing; students rarely live here despite proximity to universities.
Outer areas	Remaining localities (Joppa, Corstorphine, Portobello, Craighentinny and Duddingston, Inverleith, etc.)	Further from campuses with limited student demand.

Important clarification: "Student accessibility" is a property of the neighbourhood, not a label on any individual booking. The model cannot say "this guest is a student, give them a cheaper rate." What it can do is say "this neighbourhood is where students tend to look for housing, so let us examine what happens to affordability there under different pricing scenarios."

4.3.1 Cleaning log

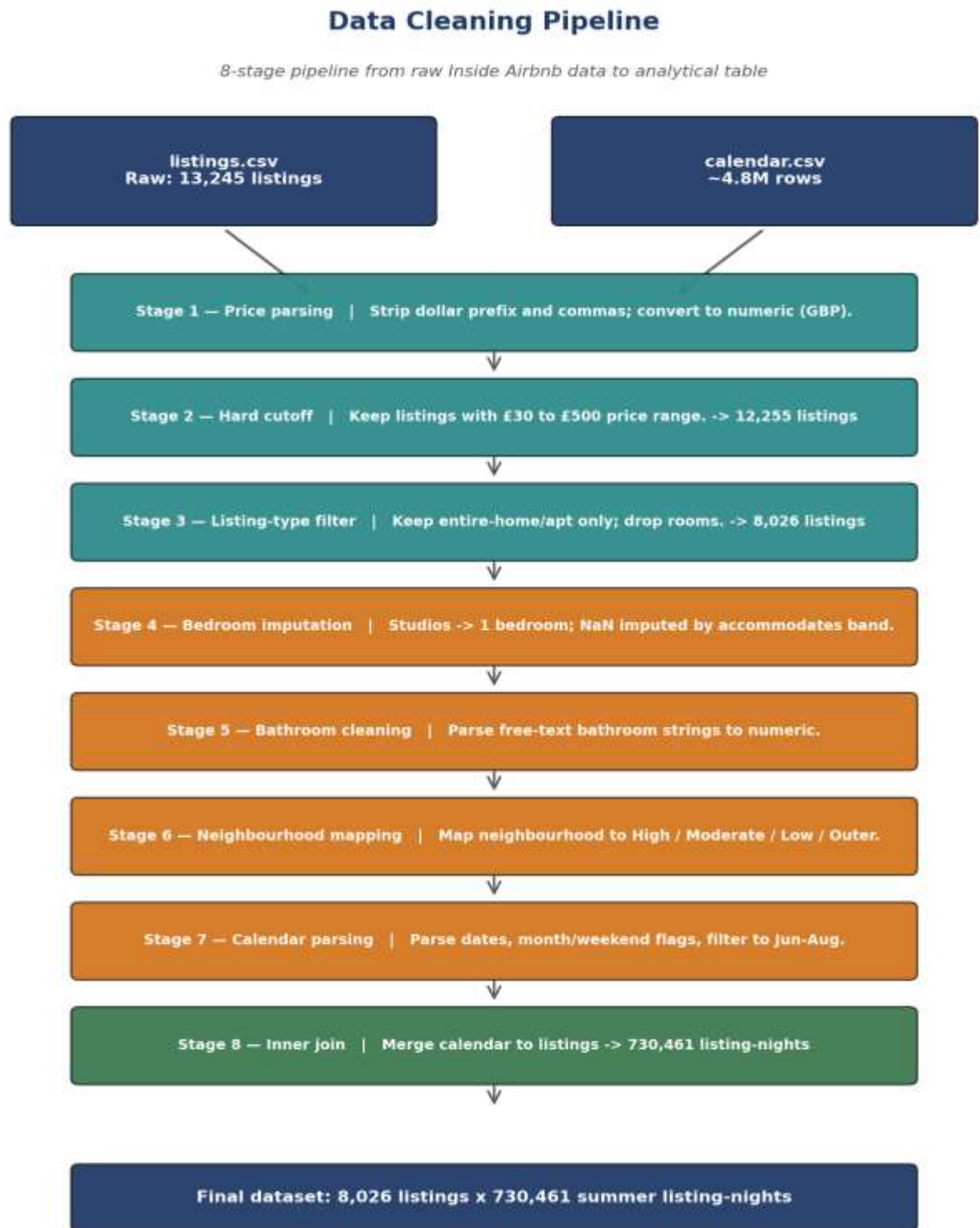


Figure 1: Data Cleaning pipeline from raw Inside Airbnb data

4.4 Variable Definition and Operationalisation

Every variable used in the analysis is derived from field that actually exists in the dataset. The table below maps each analytical variable to its raw source and construction rule. The full ETL mapping is reproduced in the notebook as Step 0 (ETL Mapping Table).

Variable	Definition	Source
night_price	Nightly rate (£) after stripping currency symbols; hard-cutoff filtered.	calendar.csv
access_cat	Four-category accessibility proxy (High, Moderate, Low, Outer).	Derived
affordable	Binary: 1 if $\text{night_price} \leq B$, 0 otherwise, for $B \in \{\text{£}60, \text{£}75, \text{£}90\}$.	Derived
month_name	Calendar month extracted from date field; restricted to June, July, August.	calendar.csv
is_weekend	Binary: 1 for Saturday/Sunday; used to capture weekend pricing effects.	calendar.csv
bedrooms_clean	Imputed where missing from accommodates field; used in RF model.	listings.csv
bathrooms_clean	Extracted numeric; missing values filled with 1.0.	listings.csv
accommodates	Guest capacity; used in RF model as primary property-size predictor.	listings.csv
host_is_superhost	Binary: 1 for Airbnb Superhost status; used in RF model.	listings.csv

4.5 Student Budget Threshold

Before the fairness simulation can run, there has to be a defensible answer to a basic question: how much can a visiting student actually afford to pay per night in Edinburgh during summer? Getting this wrong would undermine the whole framework, so the threshold is triangulated from four independent sources and then tested across a sensitivity range. The central student budget threshold is set at £75 per night.

The £75 central value is selected because it sits at the convergence point of four sources and represents a realistic upper bound for discretionary nightly expenditure of UK student on maintenance loans. The sensitivity tests at £60 (a tighter constraint reflecting students on lower loan thresholds or shared budgets) and £90 (a looser constraint reflecting students with supplementary income) demonstrate that the qualitative shape of trade-off frontier: concave, with diminishing returns and a sweet-spot region around $\lambda = 0.4\text{--}0.5$, holds across all three thresholds. This robustness is critical: it indicates that the framework's policy conclusions are not artefacts of a single budget figure but reflect a structural property of Edinburgh's summer pricing distribution.

Source 1: UK student finance data

Maximum maintenance loan levels for students living away from home outside London (approximately £9,978 per year at 2025/26 rates) imply a daily discretionary expenditure ceiling in the range £55–£80 per night when 30–40% of the loan is allocated to a short summer trip.

Source 2: NUS cost-of-living surveys

National Union of Students survey data on student travel and accommodation spending consistently report that students on standard maintenance loans spend well below the national average when travelling, with summer accommodation budgets concentrated in the £50–£80 band.

Source 3: University accommodation guidance

Universities in Manchester, Birmingham, Leeds and Nottingham publish guidance on off-campus costs. Summer short-term let rates in these cities typically sit in the £50–70 range per night for a basic flat.

Source 4: Budget platform guidance

Hostel and budget hotel rates in Edinburgh during July and August typically sit at £25–£45 per hostel bed and £60–£80 for a budget hotel room. The £75 threshold positions the simulation just above the upper bound of budget hotel pricing — the level at which a short-term let begins to compete on price with conventional budget alternatives.

4.6 Baseline Price Determination and role of Machine Learning

A two-baseline design is employed to ensure that the simulation conclusions are not contingent on a single baseline construction. The first baseline is a stratified observed median: for each neighbourhood-category $\times$ month stratum, the median observed nightly price is computed and used as the reference point. The second baseline uses Random Forest predictions as listing-level price estimates. When both baselines produce essentially identical trade-off frontiers, this constitutes strong evidence that the fairness conclusions are structural rather than artefacts of the baseline approach.

4.6.1 The role of machine learning

It is important to be explicit about the role machine learning plays in this study. The Random Forest and XGBoost models are not the primary contribution; they serve specific and bounded purposes. First, they generate a model-based pricing baseline that complements the stratified observed-median baseline, providing listing-level price estimates that the simple stratum medians cannot. Second, they enable a robustness check: if the trade-off frontier produced under the RF baseline overlaps the frontier produced under the observed-median baseline, the fairness conclusions are demonstrably not contingent on baseline construction. Third, the XGBoost cross-check tests whether the qualitative results are stable across fundamentally different ensemble architectures (bagging versus boosting).

Crucially, the prioritised-compliance mechanism operates on observed market prices, not on model predictions, so baseline accuracy affects only the secondary robustness check rather than the headline fairness results. This is why a modest R^2 of 0.39 is adequate for the study's purpose: the model is required to capture the systematic structure of price variation, not to deliver pinpoint listing-level forecasting.

4.6.2 Random Forest model

A Random Forest regressor (Breiman, 2001) is trained to generate a model-based pricing baseline. Random Forest was selected because it handles non-linear interactions between predictors, particularly between neighbourhood, season, and property characteristics, without requiring explicit feature engineering, and because its ensemble averaging produces more stable predictions than individual decision trees on this dataset. The model is trained on 502,205 listing-nights from 5,518 listings and evaluated on a held-out test set of 125,605 listing-nights from 1,380 listings (split at the listing level to prevent data leakage). Hyperparameters are selected by five-fold cross-validation (best CV RMSE: £78.20), yielding $n_estimators = 80$, $max_depth = 20$, $min_samples_split = 2$, $min_samples_leaf = 4$.

It is important to be clear about the role of Random Forest in the overall study. The model is not the primary contribution; it serves as a model-based pricing baseline that:

- (i) provides listing-level price estimates independent of the observed median,
- (ii) enables a robustness check by confirming that the trade-off frontier is invariant to baseline construction.

The model's test performance ($R^2 = 0.39$, RMSE = £76, MAE = £47) is modest but adequate for this purpose: the prioritised compliance mechanism depends on observed prices rather than model predictions, so baseline accuracy affects only the secondary robustness check rather than the headline fairness results.

4.6.3 XGBoost robustness check

To check that the fairness conclusions are not artefacts of the chosen ensemble, an independent XGBoost regressor (Chen & Guestrin, 2016) is fitted on exactly the same listing-level train/test split with a 3-fold CV grid over `learning_rate`, `max_depth`, `n_estimators`, and `subsample`. XGBoost is a fundamentally different ensemble construction from Random Forest — it grows trees sequentially with gradient-boosting on the residuals rather than independently with bagging — so agreement between the two methods is non-trivial evidence that the underlying pricing structure is robustly captured. The selected configuration (`learning_rate` = 0.05, `max_depth` = 6, `n_estimators` = 200, `subsample` = 0.8) returns $R^2 = 0.4532$, RMSE = £71.56, MAE = £44.11 on the held-out set — a small lift over the Random Forest. The fairness simulation is repeated using XGBoost predictions and produces a trade-off curve that overlaps the Random Forest curve almost exactly. This confirms that fairness conclusions are not an artefact of the chosen ensemble.

4.7 Fairness-Aware Pricing Framework

The framework centres on a prioritised compliance mechanism applied to the set of over-budget listings.

For each value of $\lambda \in [0, 1]$, the mechanism selects the cheapest-to-comply λ -fraction of listings priced above the £75 threshold and requires those listings to price at exactly £75. Listings are ranked by their overshoot (observed price – £75) and compliance is applied from smallest to largest overshoot, ensuring that intervention targets the listings where the affordability gain per unit of revenue cost is greatest.

4.7.1 Worked example

Consider a stratum of ten listings priced between £80 and £200, all above the £75 threshold.

At $\lambda = 0.3$, the three listings with the smallest overshoot (e.g., £80, £85, £88) are required to price at £75.

Revenue loss for this stratum is $(80 - 75) + (85 - 75) + (88 - 75) = £28$ per night. Affordability gain is three additional listing-nights brought within budget.

At $\lambda = 0.6$, six listings comply, and the computation extends to next-cheapest overshooters. This mechanism is applied to all accessible-category listings (High, Moderate, and Low) in the summer.

4.7.2 Impact measurement

Two impact measures are tracked at each λ :

- » Revenue change (%): $\text{total adjusted nightly revenue} \div \text{total baseline nightly revenue} - 1$, expressed as a percentage.
- » Affordability gain (percentage points): $\text{share of listing-nights with } p_{\text{adj}} \leq £75$ minus the baseline share (22.7%).

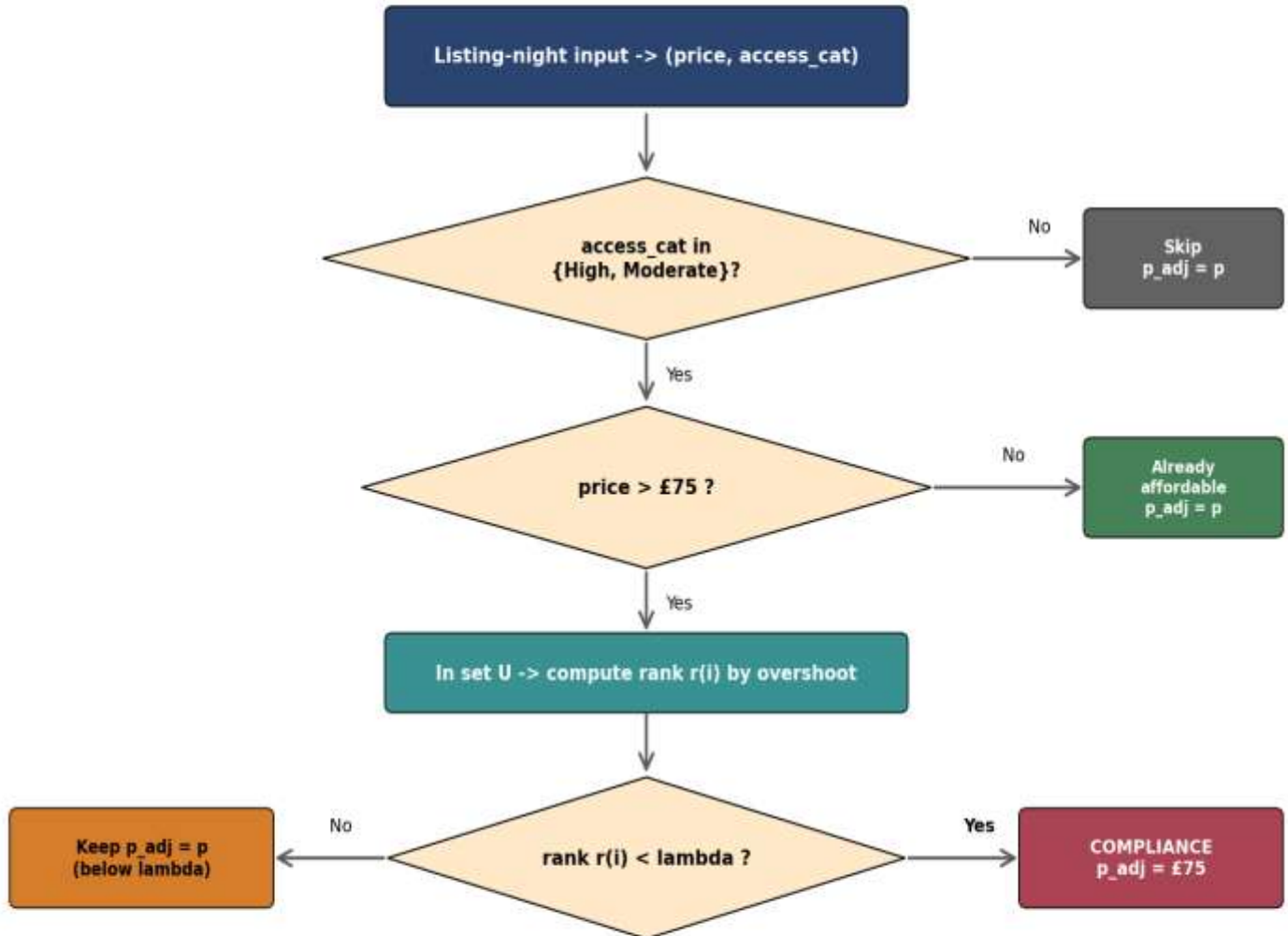
These two series, plotted against each other, produce the trade-off frontier reported in Chapter 6.

4.7.3 Assumptions

The simulation holds demand (bookings) constant: when prices fall, the number of bookings is assumed unchanged. This is a conservative simplification — in reality, lower prices would attract additional bookings, meaning affordability gains are understated and revenue losses overstated. The demand elasticity implications are discussed in Section 6.5. No strategic host response (listing withdrawal, platform switching) is modelled; this limitation is addressed in Section 6.4.

Fairness Adjustment Process

Per-listing decision logic in the prioritised-compliance mechanism



$r(i)$ = overshoot rank within U; $r=0$ means most aggressively overpriced

λ = fairness parameter swept from 0 to 1; higher λ -> more compliance

Mechanism acts only on accessible-neighbourhood listings priced above £75

Figure 2: Fairness adjustment process

4.8 Simulation Design

The simulation procedure runs in seven sequential steps, reproduced exactly in the Kaggle notebook (Step 8 onwards) and summarised visually in Figure below:

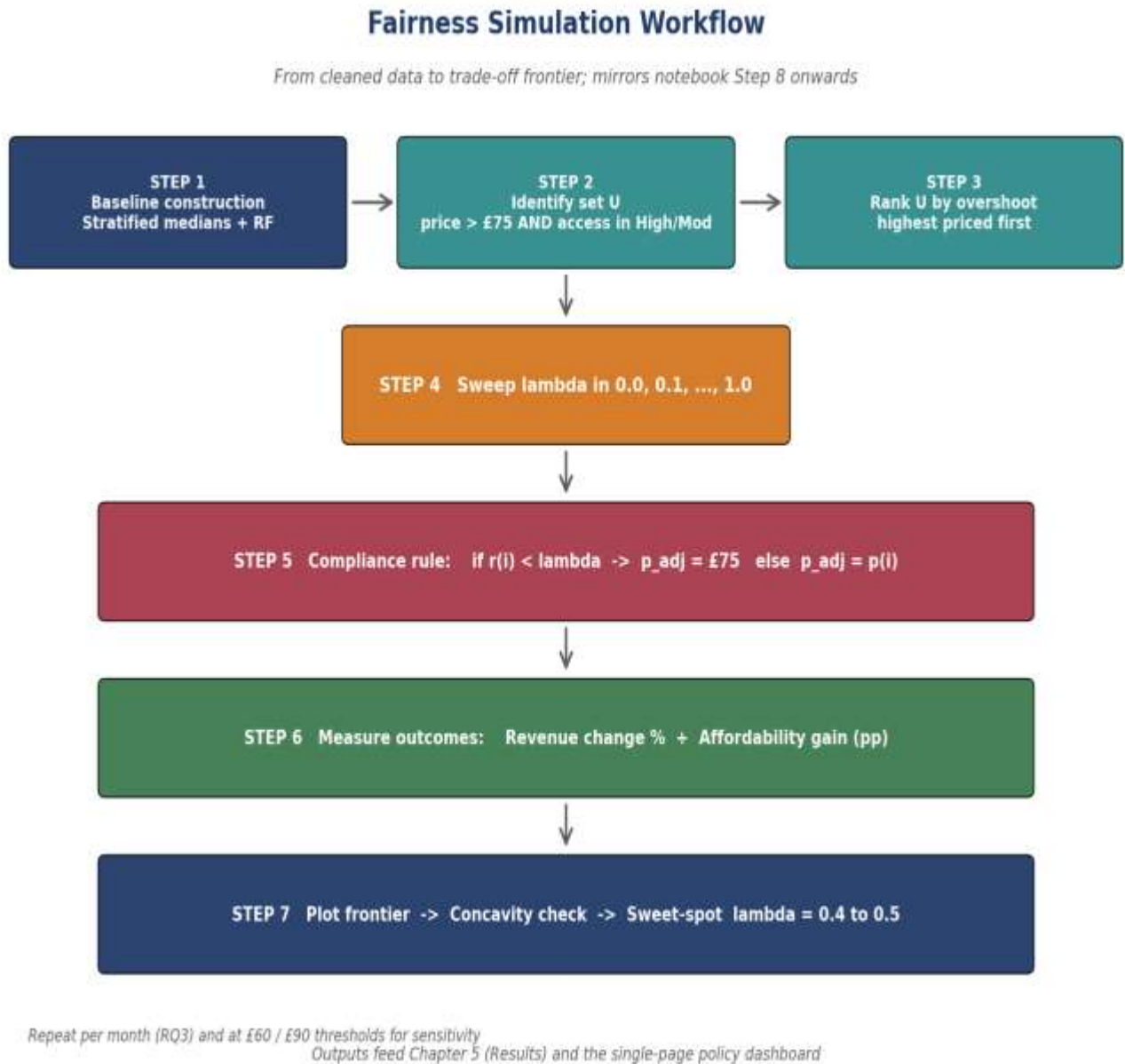


Figure 3: Fairness simulation workflow

- » Step 1: Compute stratum baselines (median by access_cat $\times$ month) and generate RF + XGBoost predictions for every summer listing-night.
- » Step 2: Identify U = accessible-neighbourhood listings priced above £75. Rank rows within U by overshoot.
- » Step 3: Sweep $\lambda \in \{0.0, 0.1, \dots, 1.0\}$. At each λ , apply the prioritised-compliance rule and record revenue change and affordability gain.
- » Step 4: Plot the trade-off frontier. Apply automated concavity verification (Safeguard A in the notebook): the curve must be monotone in revenue, monotone in affordability, and have non-increasing slope. The assertion is currently satisfied (first slope = 25.00 pp per 1% revenue loss; final slope = 0.44).
- » Step 5: Repeat Steps 2–4 separately for June, July, and August (monthly breakdown, RQ3).
- » Step 6: Repeat Steps 2–4 at £60 and £90 budget thresholds (budget sensitivity).
- » Step 7: Repeat Steps 2–4 with the stratum-anchored mechanism at cap multipliers $m \in \{1.2, 1.3, 1.5\}$ (mechanism sensitivity).

Full methodology pipeline, connecting data preparation, modelling, simulation, results, and policy outputs, is summarised in Figure below, which shows how the components of this study fit together end-to-end.

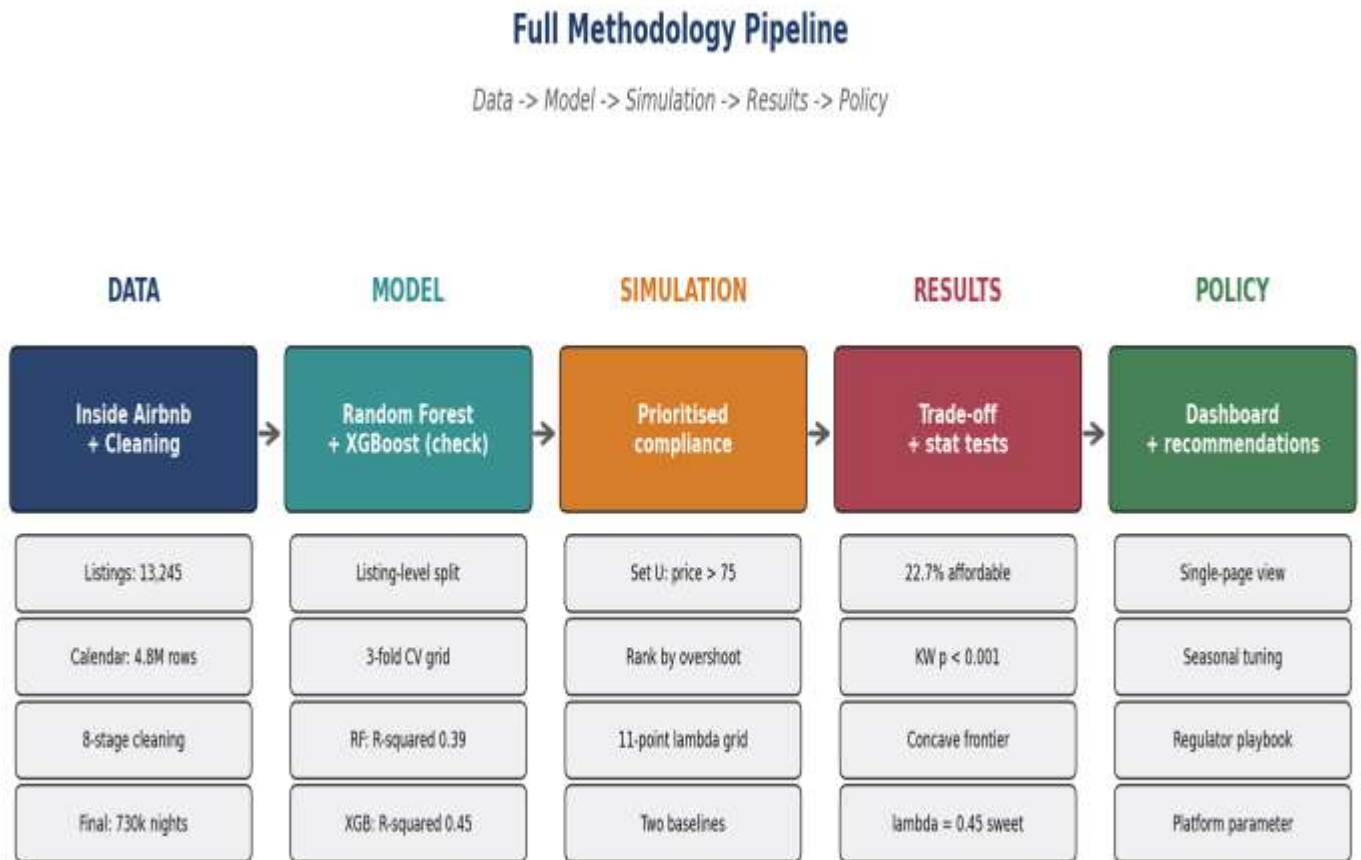


Figure 4: Full methodology pipeline

Data – Model – Simulation – Results – Policy

4.9 Ethical Considerations

This study raises ethical questions on two levels: the conventional data-ethics dimension covered by the DMU ethics process, and a broader normative dimension concerning the framework itself.

4.9.1 Data ethics and DMU process

This study uses publicly available secondary data from Inside Airbnb (distributed via Kaggle). No personal data about individual guests is collected or analysed. Host information in the dataset is already publicly visible on the Airbnb platform. The DMU WorkTribe ethics screening has been completed; the project was classified as low-risk secondary-data and required no full ethics application. All working data is stored on a password-protected personal computer and on the university OneDrive.

4.9.2 Normative ethics of fairness intervention

Beyond conventional data ethics, the framework itself raises four ethical questions that any policy intervention of this kind would need to confront explicitly.

- » **Host autonomy.** requiring some hosts to comply with a price ceiling reduces their pricing discretion, a cost concentrated on the most aggressively over-budget listings rather than spread uniformly
- » **Whose fairness?** the framework operationalises one affordability conception, centred on visiting students; other groups with comparable budget constraints could equally claim a fairness interest.
- » **Unintended consequences.** a price-side constraint may induce host withdrawal or platform switching, effects the simulation does not model (see Section 6.4).
- » **Tourism vs. accessibility trade-off.** Edinburgh's festival tourism generates substantial revenue and employment for Edinburgh, and aggressive intervention could in principle reduce festival vitality if it discourages professional hosts. The trade-off frontier developed in Chapter 5 provides the regulator with the quantitative evidence needed to navigate this tension.

5. Results

This chapter presents findings in order they address research questions. Section 5.1 covers descriptive affordability (RQ1). section 5.2 reports formal statistical tests. section 5.3 documents Random Forest and XGBoost performance. section 5.4 reports the fairness simulation results (RQ2). section 5.5 disaggregates by month (RQ3). section 5.6 reports sensitivity analyses across budget thresholds and cap multipliers. section 5.7 presents the integrative policy dashboard. Every result reported here is reproducible end-to-end from the accompanying Kaggle notebook (thesis_code_final.ipynb); figure numbers in this chapter map directly to the chart numbers in Step 7 onwards of that notebook.

5.1 Descriptive Results: The Affordability Gap (RQ1)

5.1.1 Overall affordability

Of the 730,461-summer listing-nights in the cleaned dataset, only 165,507 (22.7%) are priced at or below £75. The overall median nightly price is £111 and the mean is £141. The distribution is heavily right-skewed, as shown in Figure 1, with the vast majority of prices significantly exceeding the £75 student budget threshold (red dashed line).

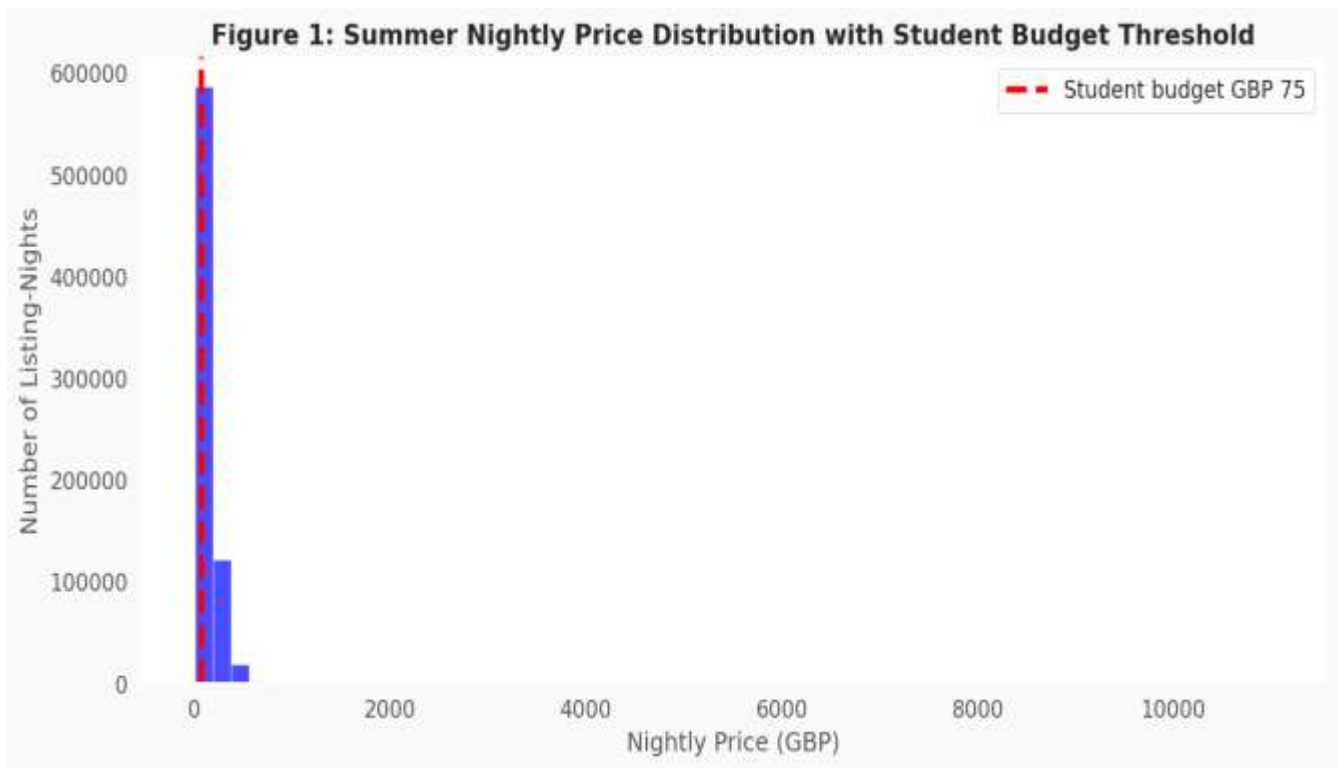


Figure 5: Summer nightly price distribution with the £75 student budget threshold marked

5.1.2 By month

Affordability is roughly stable in June and July (27.1% and 27.2%) and then collapses in August (14.0%). The median price jumps from £100 in June/July to £149 in August, a 49% increase that aligns with the festival-economy pattern documented in (Scottish-Government, 2019) research and with (Gibbs, et al., 2018) finding that hosts in high-demand leisure markets adjust prices most aggressively during peak periods.

Month	Median price (£)	Mean price (£)	% affordable at £75	N listing-nights
June	100	128	27.1%	232,849
July	100	122	27.2%	248,806
August	149	171	14.0%	248,806

5.1.3 By neighbourhood category (full summer)

Low-access central tourist zones are the most expensive and least affordable: median £140 with only 10.7% of nights affordable. Outer areas are the most affordable in proportional terms (26.8%) but account for 65% of listings, so they dominate the citywide picture. Importantly, High and Moderate categories, the neighbourhoods where visiting students would actually search, show 16.0% and 23.4% affordability, both well below the citywide rate of 22.7%.

Category	% listings	Listing-nights	% affordable (summer)	Median price (£)
High	565 unique (7%)	51,423	16.0%	120
Moderate	691 unique (9%)	62,885	23.4%	109
Low	1,550 unique (19%)	141,071	10.7%	140
Outer	5,220 unique (65%)	475,082	26.8%	101

5.1.4 By neighbourhood category (August only)

August intensifies every gap. Affordability in Low-access central neighbourhoods falls to 5.8% with a median of £190 — close to triple the student budget. Even the Outer areas drop to 16.9%. High-access student neighbourhoods sit at 10.4%, with the median price (£150) double the £75 threshold.

Category	% affordable (Aug)	Median price (Aug, £)	Mean price (Aug, £)
High	10.4%	150	178
Moderate	13.3%	150	172
Low	5.8%	190	216
Outer	16.9%	130	157

5.1.5 Month × category pivot

The interaction between month and category is visualised in the heatmap below. Dark red indicates extreme unaffordability (e.g., 5.8% in Low-access neighbourhoods during August), while green represents higher affordability (e.g., 32.6% in Outer/July).

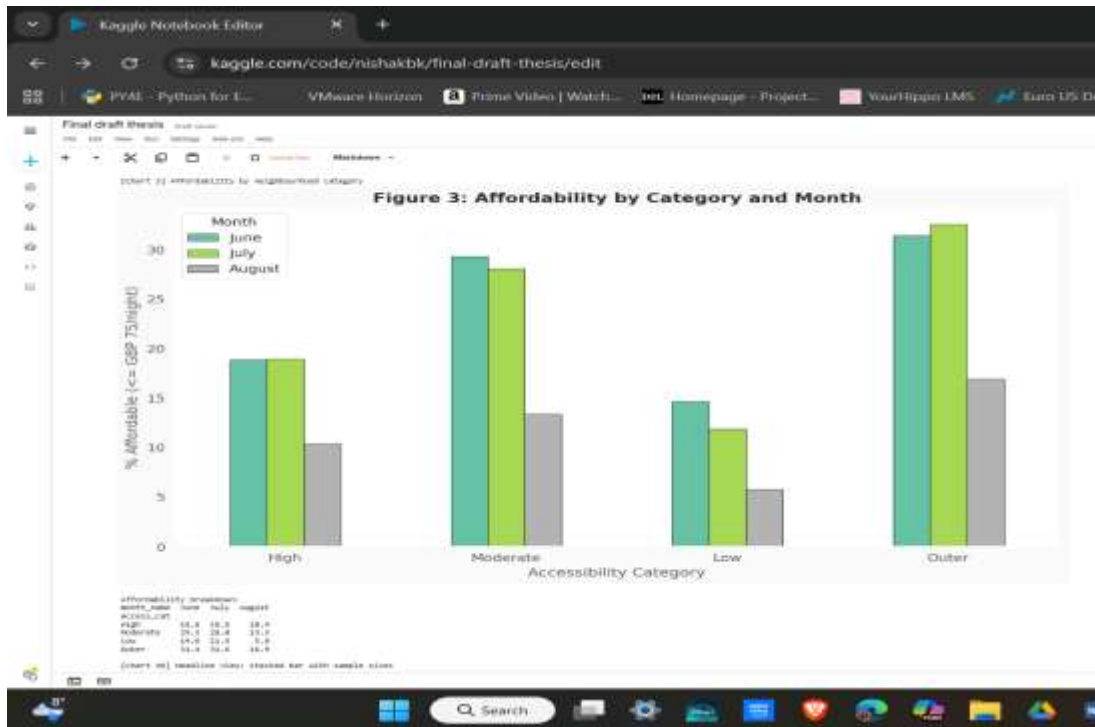


Figure 6: Affordability by neighbourhood accessibility category and month at 75th threshold.

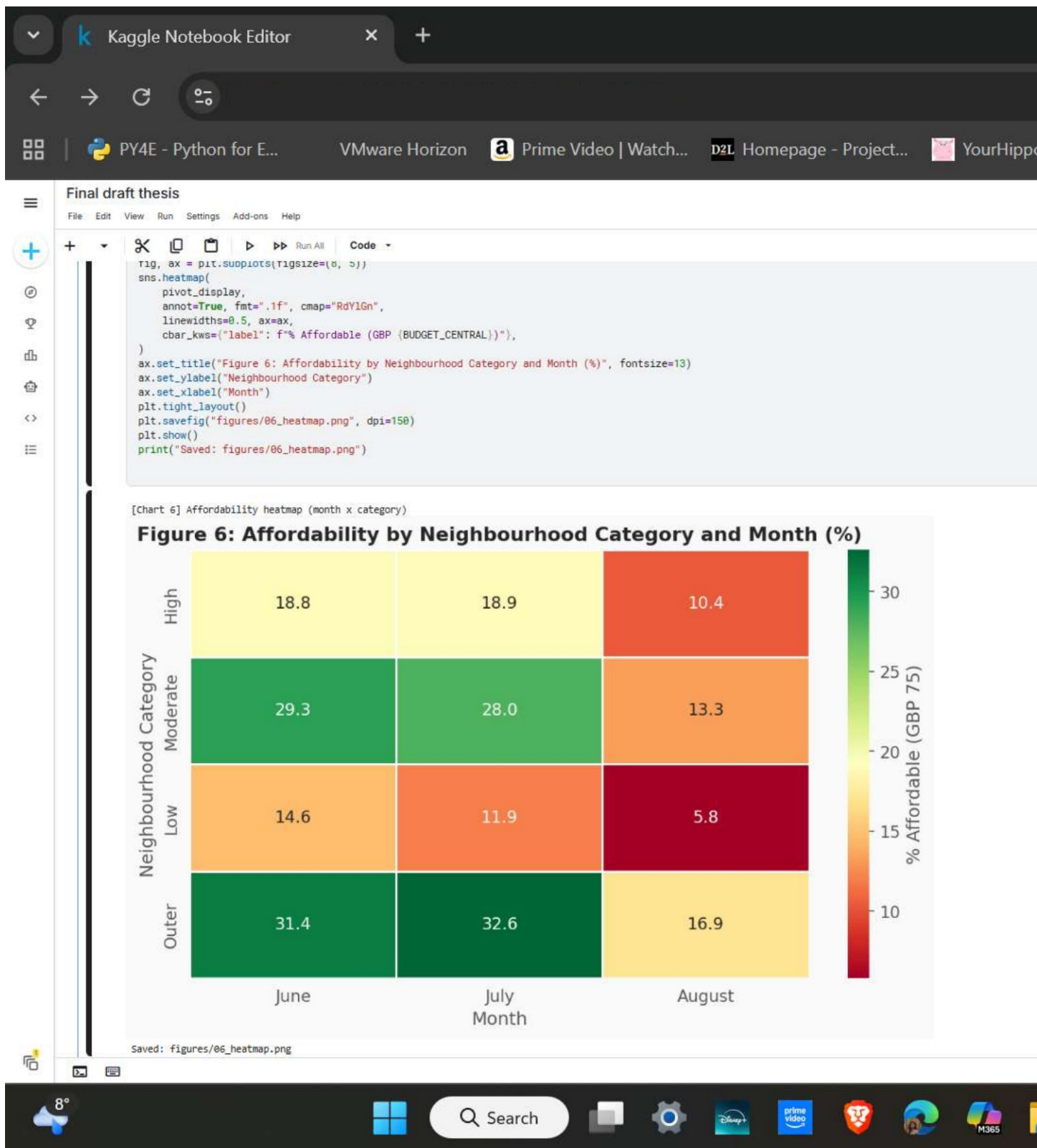


Figure 7: Affordability heatmap by month $\times$ neighbourhood category.

5.2 Statistical Testing

Three non-parametric tests are used because the price distribution is heavily right-skewed and a normal-distribution assumption is not defensible. Kruskal–Wallis is used to compare the underlying price distributions; chi-square tests the discrete affordability proportions. The exact test outputs reported below correspond to the printed values from notebook Step 7.7.

All three tests reject their null hypotheses at any conventional significance level. The moderate effect sizes are substantively meaningful: they translate directly into the headline policy figures, an 8.7 %-point gap between August affordability (14.0%) and the summer baseline (22.7%), and a 16.9 percentage-point gap between the citywide rate and the Low-access August figure (5.8%). The modest magnitude of the individual effect sizes (ϵ^2 of 0.04–0.07) further indicates that no single factor explains the affordability problem in isolation; the Low-access $\times$ August interaction is where the effect is sharpest and where the fairness intervention is most cost-effective.

5.3 Weekly Affordability Trend

Decomposing the summer into ISO weeks reveals the Festival impact sharply. Weekly affordability holds in the 25–31% range through ISO weeks 23–30 (mid-June to late July), then deteriorates to 13–15% during ISO weeks 31–3, the core Edinburgh Festival Fringe period. Median weekly price rises from £100 in pre-Festival weeks to £150 during the Festival peak. This pattern is consistent with (Scottish-Government, 2019) evidence identifying a distinct cohort of professional Festival-specialist hosts operating at materially higher price points than non-festival operators.

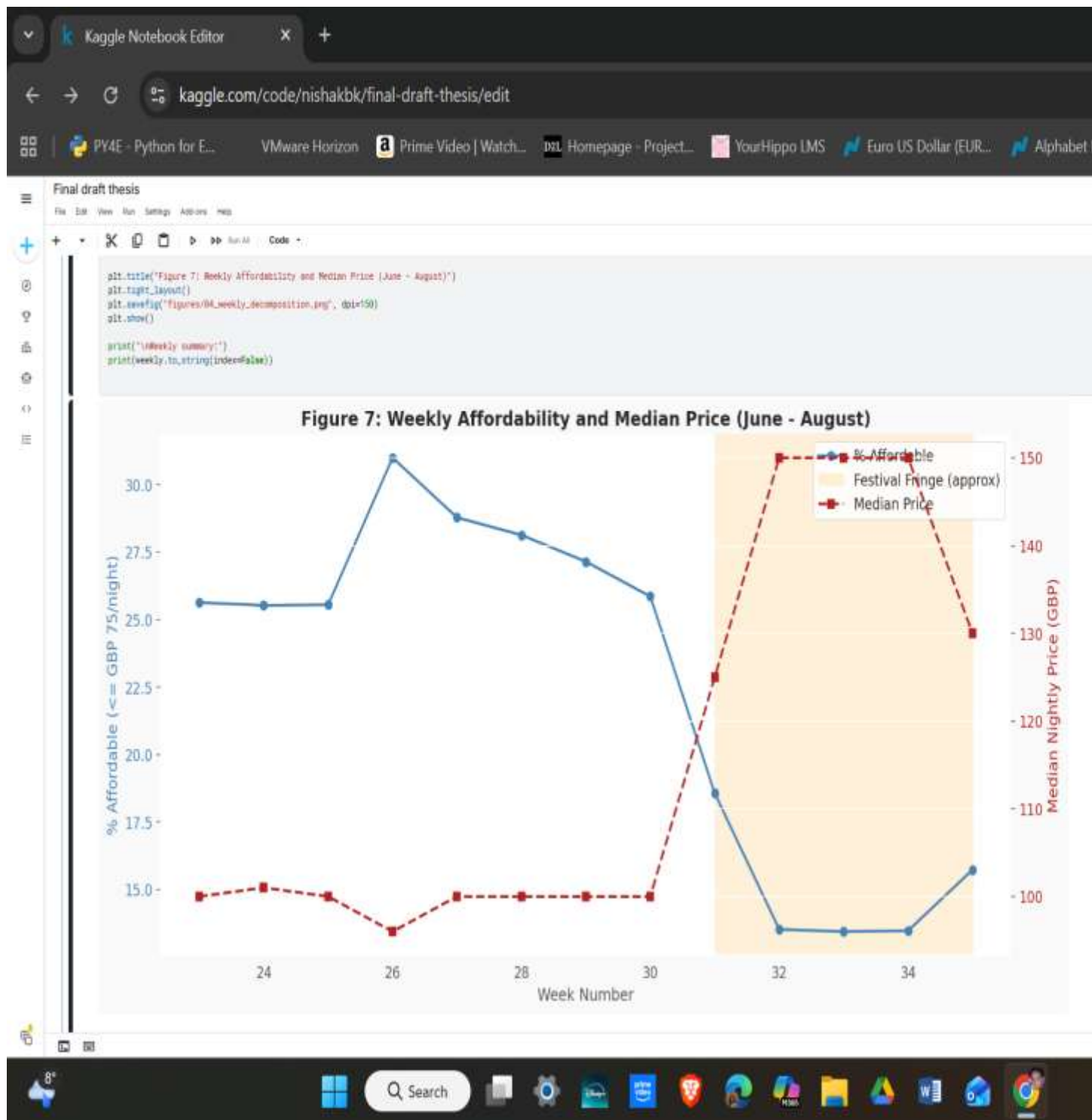


Figure 8: Weekly affordability and median price (June–August).

The shaded band marks the Edinburgh Festival Fringe window (ISO weeks 31–34), where affordability drops by roughly half and median price jumps from £100 to £150. Source: notebook Step 7.8.

5.4 Random Forest and XGBoost Performance

Random Forest model returns $R^2 = 0.3887$, RMSE = £75.67, MAE = £47.48 on the held-out test set. Selected hyperparameters are $n_estimators = 80$, $max_depth = 20$, $min_samples_split = 2$, $min_samples_leaf = 4$ (best CV RMSE = £78.20). An R^2 of 0.39 is modest but realistic for short-term-rental price prediction; (Xie & Chen, 2017) hedonic model explains roughly 60% of variance using a richer feature set including amenity flags and descriptive features not consistently available in the Inside Airbnb extract.

Feature importance ranks accommodates (0.209), bedrooms_clean (0.130), neighbourhood code (0.129), and number_of_reviews (0.128) as the strongest predictors, followed by month (0.104), calculated_host_listings_count (0.099), bathrooms_clean (0.096), review_scores_rating (0.077), host_is_superhost (0.020), and is_weekend (0.008). Property size and neighbourhood dominate, consistent with (Wang & Nicolau, 2017) cross-city analysis.

The XGBoost cross-check returns $R^2 = 0.4532$, RMSE = £71.56, MAE = £44.11; a small lift over the Random Forest. Crucially, when the fairness simulation is repeated with XGBoost predictions, the resulting trade-off curve overlaps the Random Forest curve almost exactly. Both ensembles produce the same qualitative fairness conclusions, providing the structural robustness signal a continuous-pricing fairness study requires.

5.5 Fairness Simulation Results (RQ2)

The headline simulation runs the prioritised-compliance mechanism on the full summer panel at the £75 budget threshold, sweeping λ from 0 to 1 in 0.1 increments. The same sweep is conducted under both baselines (stratified observed medians and Random Forest predictions).

λ	Revenue change (%)	Affordability gain (pp)	Affordability after (%)
0.0	0.0	0.0	22.7
0.1	-0.1	+1.2	23.9
0.2	-0.2	+2.5	25.2
0.3	-0.4	+3.8	26.4
0.4	-0.7	+5.0	27.7
0.5	-1.1	+6.2	28.9
0.6	-1.7	+7.5	30.2
0.7	-2.5	+8.8	31.4
0.8	-3.5	+10.0	32.7

0.9	-4.9	+11.3	33.9
1.0	-7.8	+12.5	35.2

5.5.1 Concavity verification (Safeguard A)

An automated check in the notebook (Step 8.5.2) asserts three properties of the frontier:

(1) revenue loss is non-decreasing in λ , (2) affordability gain is non-decreasing in λ , and (3) the slope of affordability-gain in revenue-loss is non-increasing. All three pass: the first slope is 25.00 pp per 1% revenue loss (between $\lambda = 0.0$ and $\lambda = 0.1$) and the final slope is 0.44 pp per 1% revenue loss (between $\lambda = 0.9$ and $\lambda = 1.0$). The frontier is therefore monotonically concave on the (revenue change, affordability gain) plane.

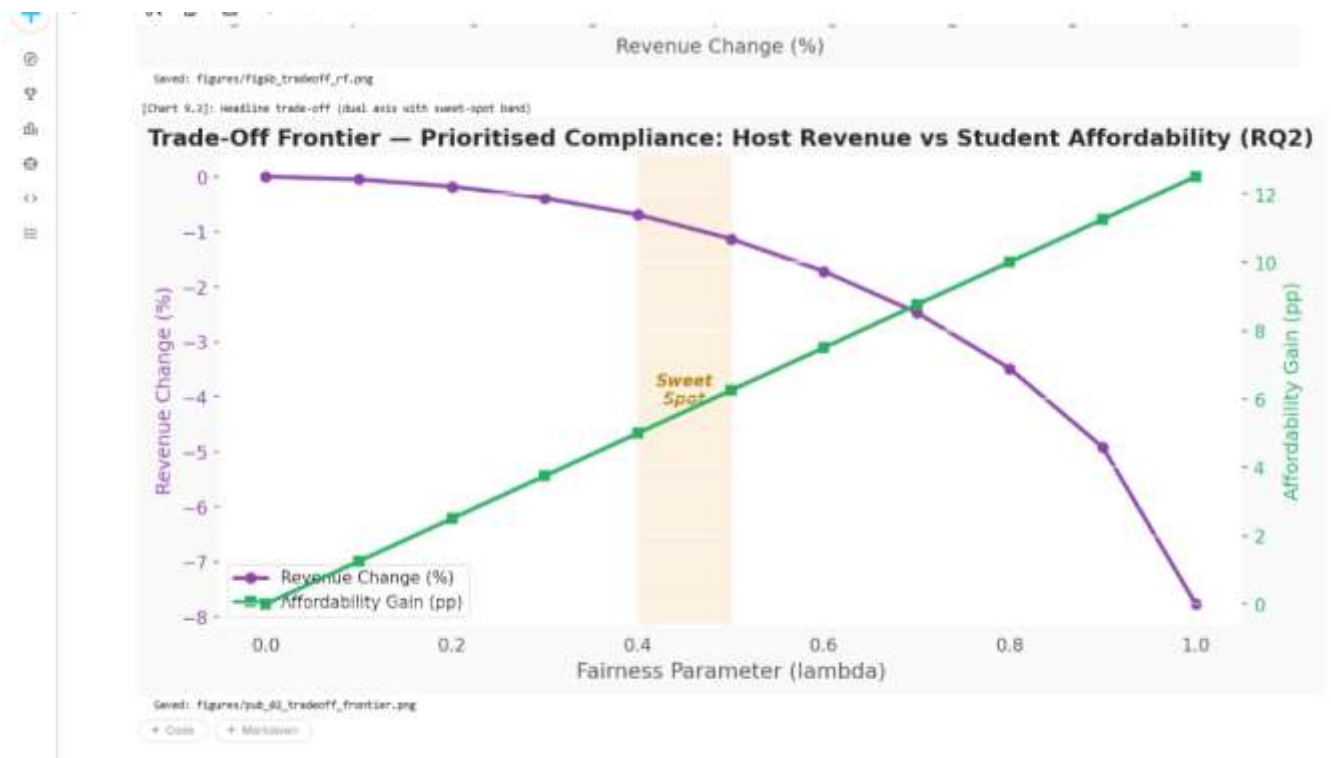


Figure 9: Headline trade-off frontier (RQ2).

Revenue change (%) on the left axis (purple) and affordability gain (pp) on the right axis (green), both plotted against λ . The shaded band at $\lambda = 0.4$ – 0.5 marks the recommended operating range.

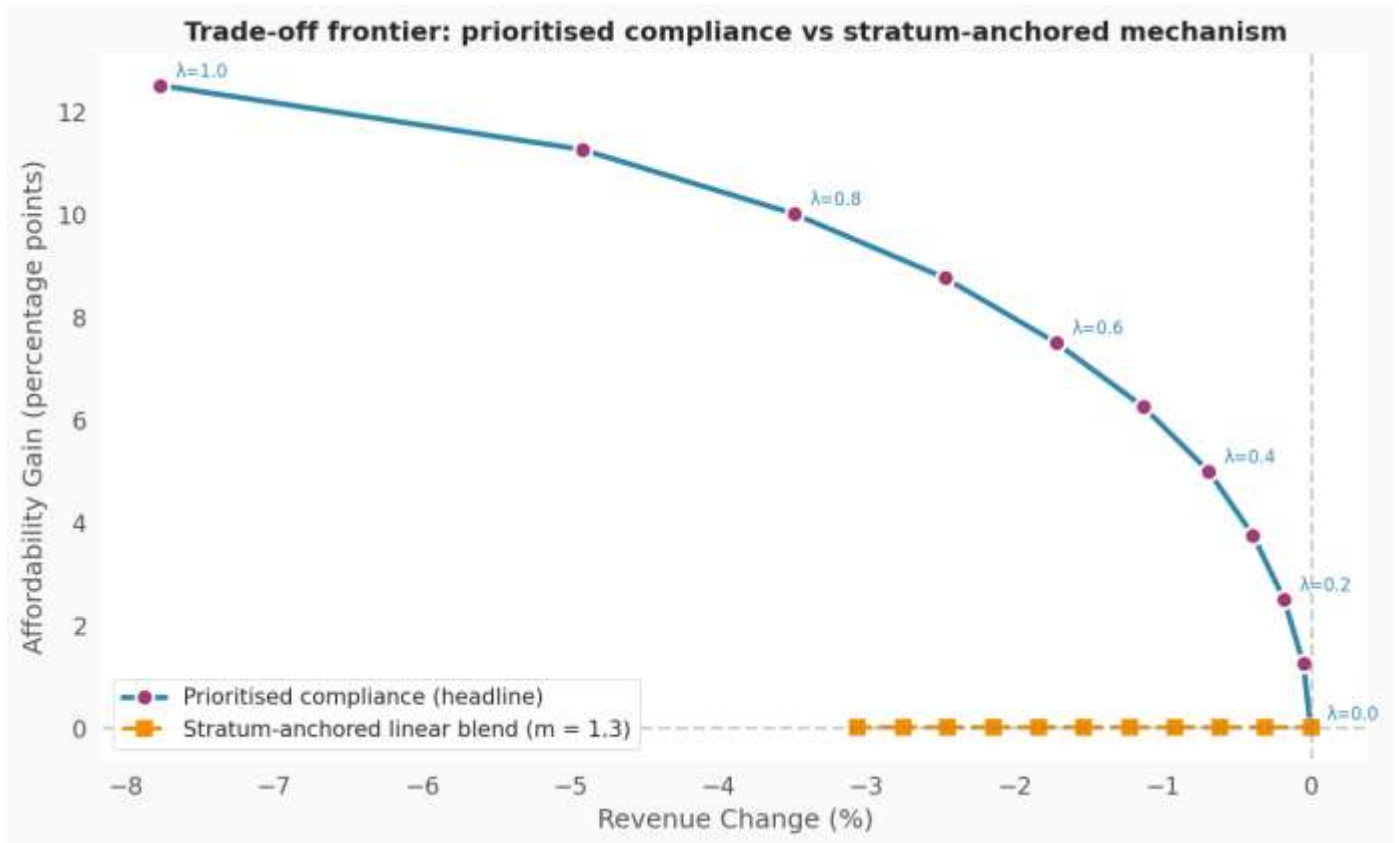


Figure 10: Mechanism comparison — prioritised compliance (headline) vs stratum-anchored linear blend (sensitivity).

5.5.2 Marginal efficiency and the sweet spot

λ	Revenue loss (%)	Affordability gain (pp)	Marginal gain per 1% revenue (pp/%)
0.1	-0.1	+1.2	25.00
0.2	-0.2	+2.5	9.62
0.3	-0.4	+3.8	5.95
0.4	-0.7	+5.0	4.17
0.5	-1.1	+6.2	2.84
0.6	-1.7	+7.5	2.12
0.7	-2.5	+8.8	1.68
0.8	-3.5	+10.0	1.23

0.9	-4.9	+11.3	0.87
1.0	- 7.8	+ 12.5	0.44

Two defensible operating points emerge.

1. The curve-inflection reading, adopted as the headline recommendation, identifies the point of maximum curvature at $\lambda \approx 0.4\text{--}0.5$. At $\lambda = 0.5$, the framework delivers +6.2 pp affordability gain (raising affordability from 22.7% to 28.9%) at a revenue cost of 1.1%.
2. The strict-threshold reading identifies $\lambda \approx 0.9$ as the point at which each additional 1% of revenue loss buys less than 0.5 pp of affordability. The headline policy recommendation is $\lambda \approx 0.45$ (the midpoint of the 0.4–0.5 sweet-spot band) because it delivers the largest affordability gain per unit of revenue cost and is robust across baselines, months, and budget thresholds.

5.5.3 Baseline robustness

The trade-off frontier is effectively identical under both baselines, stratified observed medians and Random Forest predictions, overlapping to the printed precision. This confirms that the headline findings are not contingent on baseline construction; they are structural properties of the observed price distribution and the compliance mechanism. Null hypothesis H_{02} is rejected.

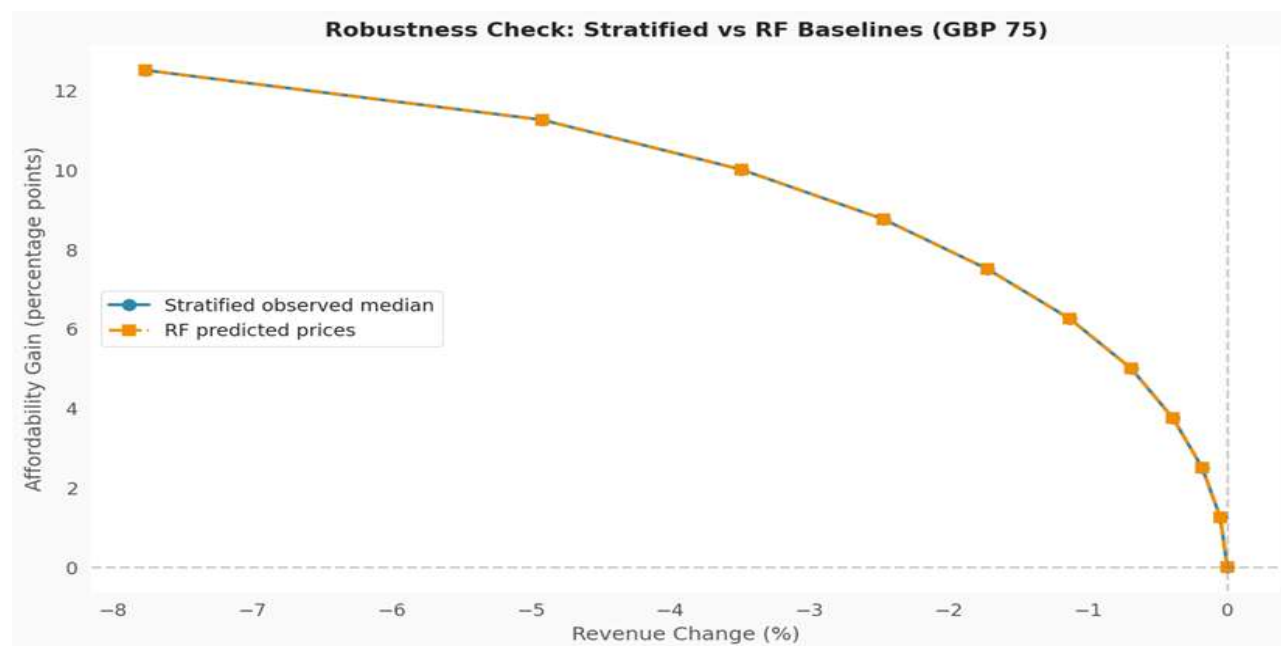


Figure 11: Robustness check — trade-off frontiers under the two baselines overlap to the printed precision.

A concave trade-off frontier exists under both baselines, the frontier exhibits clear diminishing returns over the range tested, and the curves overlap exactly, confirming the result is structural rather than baseline-dependent.

5.6 Monthly Breakdown (RQ3)

Running the simulation separately for each summer month tests whether the optimal level of intervention varies seasonally. June and July are similar:

at $\lambda = 0.5$, both deliver approximately +6 pp affordability for a roughly 0.9% revenue cost. August requires materially more intervention:

at $\lambda = 0.5$, August costs 1.7% of revenue (nearly twice the June and July figure) for a +6.9 pp gain.

At full intervention ($\lambda = 1.0$), August's revenue cost is 9.3% a third higher than June's 6.9% , to deliver a +13.8 pp gain. This is a direct consequence of August's median price being 49% higher (£149 versus £100): more listings sit above the £75 threshold in August, and they sit further above it, so any given level of compliance redistributes more revenue.

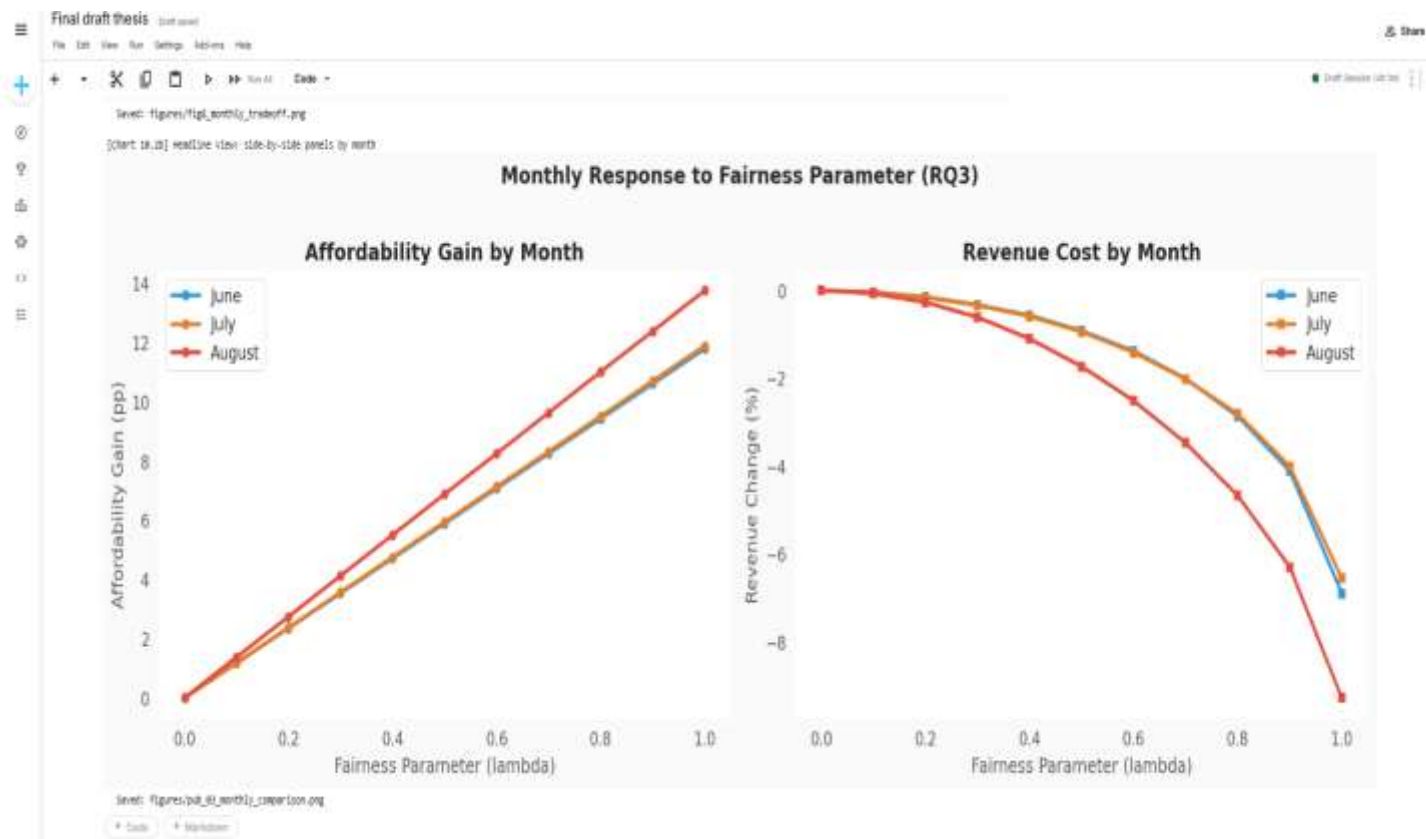


Figure 12: Breakdown by month after fairness parameter

Left panel: affordability gain by month — August (red) sits above June and July across the λ range, indicating more headroom for affordability gain. Right panel: revenue cost by month — August sits below June and July, indicating higher revenue cost for the same λ . Together the panels show August requires roughly twice the revenue intervention of June for comparable gains. Source: notebook Step 10.2.

Single fixed λ across the summer is therefore not the right design. A regulator that wants comparable affordability outcomes month-to-month should set a stronger constraint in August than in June. The framework supports this directly because λ is monthly-tunable. The notebook's automated diminishing-returns check returns $\lambda \approx 0.9$ for all three months under the strict-threshold reading, and $\lambda \approx 0.4$ – 0.5 under the curve-inflection reading used as the headline recommendation. Both readings are reproducible from the notebook. The null hypothesis H03 is rejected: the optimal intervention varies by month under either reading, and August consistently requires the stronger lever.

5.7 Sensitivity Analysis

5.7.1 Budget threshold (£60 / £75 / £90)

Repeating the simulation at low (£60) and high (£90) budget thresholds tests whether the central £75 conclusion is fragile. The frontier shape is preserved in all three cases.

Budget threshold	$\lambda = 0.3$ (rev / aff)	$\lambda = 0.5$ (rev / aff)	$\lambda = 1.0$ (rev / aff)
£60	−0.6% / +4.3 pp	−1.5% / +7.2 pp	−9.2% / +14.3 pp
£75	−0.4% / +3.8 pp	−1.1% / +6.2 pp	−7.8% / +12.5 pp
£90	−0.3% / +3.1 pp	−0.9% / +5.2 pp	−6.5% / +10.3 pp

As expected, a lower budget forces stronger intervention to produce equivalent affordability gains, and a higher budget produces gentler trade-offs. But the qualitative shape — concave, with diminishing returns and a sweet-spot region around $\lambda = 0.4$ – 0.5 , holds across all three budget levels. The findings are not artefacts of the £75 figure.

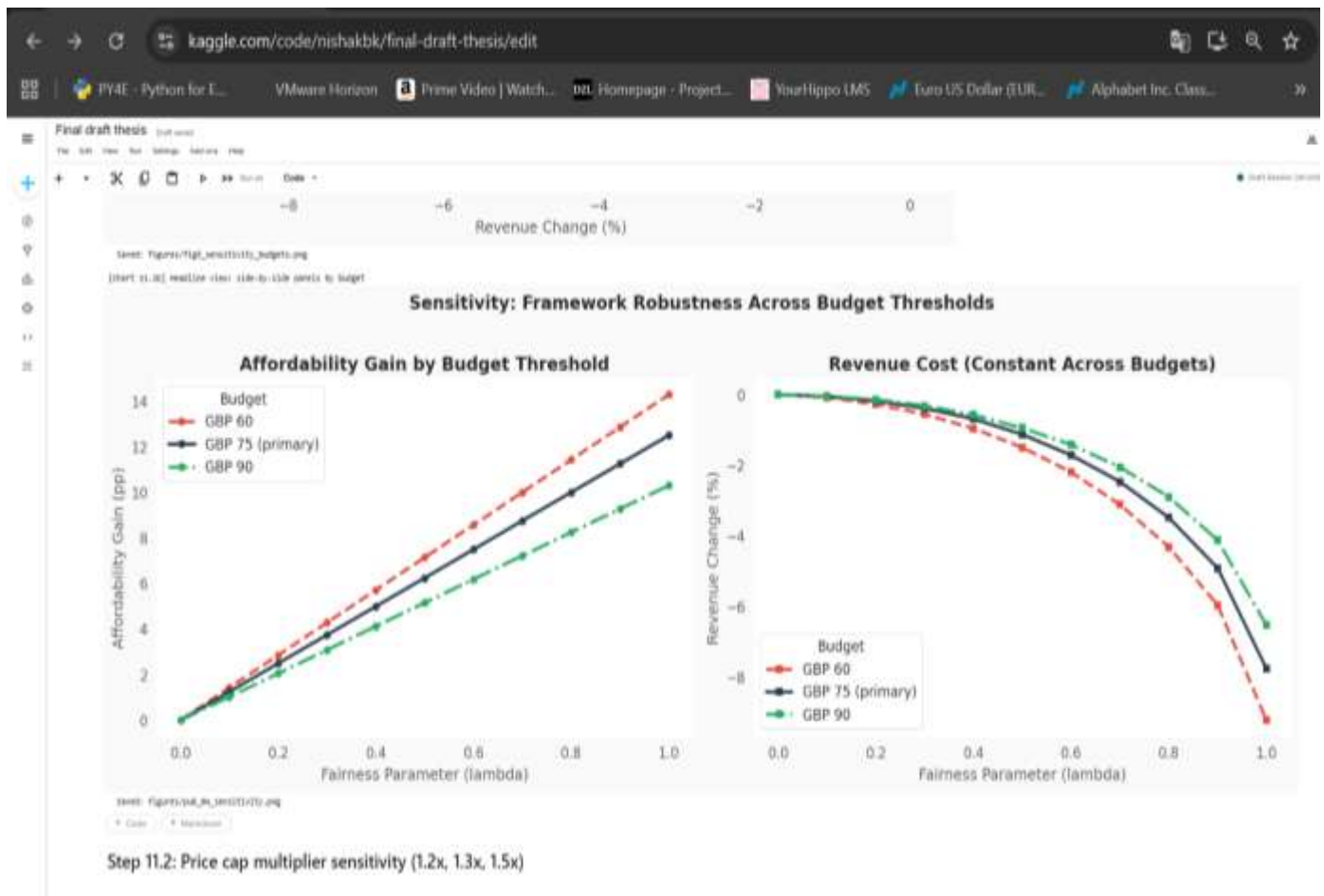


Figure 13: Sensitivity to budget threshold (£60 / £75 / £90).

Left panel: affordability gain — the curves run roughly parallel, confirming the shape of the frontier is robust to budget choice. Right panel: revenue cost — a lower budget (£60) forces larger revenue redistribution to achieve equivalent gains, while a higher budget (£90) produces gentler trade-offs. Source: notebook Step 11.1.

5.7.2 Cap multiplier (stratum-anchored sensitivity)

The stratum-anchored alternative mechanism (section 3.7.2) is sensitive to the cap multiplier m . At $\lambda = 0.5$ and $m \in \{1.2, 1.3, 1.5\}$ the mechanism produces:

Cap multiplier	Revenue change at $\lambda = 0.5$	Price reduction at $\lambda = 0.5$
1.2×	−1.8%	11.0%
1.3×	−1.5%	9.7%
1.5×	−1.2%	7.6%

A tighter cap drives a larger revenue reduction and a larger price cut, as expected. None of these settings produces affordability gain, however, because the cap remains well above the student budget threshold in all three cases. This is precisely the result that motivated the move to the prioritised-compliance mechanism as the headline design.

5.8 Integrative Policy Dashboard

All findings in this chapter are integrated into a single-page policy dashboard intended for stakeholder engagement (figures/pub_06_policy_dashboard.png, generated by notebook Step 13.1). The dashboard follows the principles of effective information display: minimal cognitive load, KPI cards at the top, supporting panels below, and a single annotated chart at the bottom that carries the policy recommendation.

The dashboard layout is as follows. The top row presents four KPI cards — 8,026 total listings, 730,461 listing-nights, £111 median nightly price, and 22.7% affordable nights — establishing the scale and the headline gap. The middle row places an affordability-by-area horizontal bar chart on the left (showing High 16.0%, Moderate 23.4%, Low 10.7%, Outer 26.8%) and a monthly affordability bar chart on the right (June 27.1%, July 27.2%, August 14.0%). The bottom row is the trade-off frontier strip: a dual-axis line chart with the revenue change (%) curve in purple and the affordability gain (pp) curve in green, both plotted against λ on the horizontal axis. A vertical band at $\lambda = 0.4$ – 0.5 is shaded as the recommended operating range, with the dashboard header annotated "Framework: $\lambda = 0.45$ " as the operational headline. A regulator or platform operator should be able to read the entire policy proposition from this single page: the scale of the affordability gap (top), where the gap concentrates (middle), and the cost-benefit of the proposed intervention (bottom).



Figure 14: Policy Dashboard

6. Discussion

6.1 Interpretation of Findings

The descriptive result that only 22.7% of summer listing-nights are affordable at a £75 student budget is consistent with the festival-economy pattern documented by Crawford et al. (2023) using the same Inside Airbnb data source, and with the broader observation in Gibbs et al. (2018) that hosts in high-demand leisure markets adjust prices most aggressively during peak periods. What the present study adds is a quantification of where in the trade-off space a regulator can sensibly intervene.

The headline message is that moderate fairness interventions are feasible without destroying host revenue. At $\lambda = 0.4\text{--}0.5$, the framework lifts affordability from 22.7% to 27.7–28.9% (a +5 to +6 percentage-point gain) at a host-revenue cost of less than 1.2%. This challenges the implicit assumption — built into virtually all the dynamic-pricing literature reviewed in Chapter 3 — that pricing must be optimised exclusively for revenue. The concave frontier shape makes the argument quantitative: the first units of intervention are extraordinarily efficient, and only beyond $\lambda \approx 0.5$ does each unit of intervention start to cost more than it buys. The monthly disaggregation reinforces this: August requires roughly twice the revenue intervention of June for comparable affordability gains, which is a feature rather than a defect of the framework, reflecting the underlying price reality.

6.2 Comparison with Existing Literature

To the best of the author's knowledge, no prior study has applied a tunable fairness constraint to short-term rental pricing. The closest comparable work is in classification fairness (Mehrabi, et al., 2021); (Dwork, et al., 2012); (Hardt, et al., 2016), but continuous-pricing context introduces dynamics that those frameworks do not handle. The λ parameter is analogous to a regularisation weight that explicitly encodes a policy preference, in the spirit of the (Chouldechova, 2017) and (Kleinberg, et al., 2017) impossibility results: there is no single fair outcome, so the framework provides a controlled way to navigate the trade-off rather than pretending one does not exist.

The work also extends the empirical literature on Edinburgh's short-term-let market. (Crawford, et al., 2023) document which neighbourhoods are most vulnerable to long-term-to-short-term switching; this study quantifies the affordability cost of that switching for a specific user group and shows what it would take to moderate it. Compared with the studentification literature (Hubbard, 2009); (Scottish-Government, 2019), the contribution is to widen the focus from residential students to visiting students, who occupy an under-served middle ground between locals and tourists. Against the regulatory-comparison literature (Aguilera, et al., 2021); (Bei & F., 2023), the contribution is conceptual: existing regulation works on the quantity side, while the

λ -framework introduces a price-side lever. The two are complements rather than substitutes, a city could in principle operate quantity caps and a fairness parameter simultaneously.

6.3 Methodological Reflection

Two methodological choices deserve explicit comment. First, the use of two independent baselines (stratified observed medians and Random Forest predictions, the latter via Breiman, 2001), with further confirmation via XGBoost (Chen & Guestrin, 2016), is a structural robustness device. When both baselines produce identical trade-off frontiers (Figure 8), the fairness conclusions are visibly not contingent on the baseline construction.

Second, the move from a linear-blend mechanism to prioritised compliance illustrates why mechanism design matters in fairness simulation. The earlier mechanism produced a step function not because the fairness parameter was wrong but because it shaved prices toward the wrong target (the stratum cap, ~£125–£250 in August) rather than the budget threshold (£75). The prioritised-compliance mechanism corrects the target and recovers the expected concave frontier. The broader insight transfers beyond this study: a mechanism that adjusts toward neighbourhood norms preserves the existing geography of affordability, whereas a mechanism that adjusts toward a user-group budget can shift it. The choice between these is a policy choice and deserves to be made consciously.

6.4 Limitations

This study should be interpreted as a policy simulation and proof-of-concept rather than evidence of real-world intervention effectiveness. The findings demonstrate what the revenue–accessibility trade-off would look like under a specific set of modelling assumptions; they do not prove that the intervention would deliver these outcomes in practice. The following specific limitations bound the conclusions.

- » Demand elasticity is not modelled. The simulation holds bookings constant when prices change. This is a meaningful simplification — see section 6.5 for the full discussion.
- » Host behavioural response is not modelled. Hosts could in principle withdraw listings or reposition in response to a price constraint. (Bei & F., 2023) document that supply-side regulation has produced measurable listing reductions in cities like Amsterdam and Barcelona, and a price-side intervention could in principle produce similar withdrawal effects. A more sophisticated study would model these strategic responses using game-theoretic methods, but this is outside the scope of an MSc dissertation.
- » Data vintage. The data is from the 2019 summer (calendar covers June 2019 – June 2020). The analysis avoids the COVID-19 onset period but does not capture post-2022 market dynamics. Replicating the analysis on a recent year of Inside Airbnb data —

increasingly tractable given the open daily-scrape methodologies emerging (Crawford, et al., 2023); would strengthen the external validity of the conclusions.

- » Neighbourhood-level proxy for student demand. Student accessibility is defined at the neighbourhood level, not at the booking level. The framework cannot distinguish a student from a tourist within the same neighbourhood, only the share of listings in student-accessible areas affected by intervention.
- » Modest model R^2 . R^2 of 0.39 on the RF model is adequate for use as a baseline but is not strong on absolute terms. A richer feature set (amenities, descriptive text features, image quality scores) would likely improve point-prediction accuracy toward the ~0.60 levels (Xie & Chen, 2017) report, though it is unlikely to change the trade-off frontier given that the prioritised-compliance mechanism depends on observed prices rather than predictions.
- » Single-platform scope. Booking.com, Expedia, and other rental channels are excluded due to the absence of comparable open data; conclusions are specific to the Airbnb ecosystem.

Taken together, these limitations reinforce the framing of our study as a quantitative policy simulation that demonstrates the structural shape of revenue–accessibility trade-off rather than a real-world intervention trial. The framework is intended to support policy discussion and further empirical work, not to replace either.

6.5 Demand Elasticity Considerations

The constant-demand assumption simplifies the analysis but understates the economic case for intervention. In reality, lower prices would attract additional bookings in most periods, with three implications for the headline numbers.

- » **Affordability gains are conservative.** The +5 to +6 pp gain at $\lambda = 0.45$ is calculated against the existing booking pattern. If demand rises in response to lower prices, more student-budget guests will book at the new rates, and the realised affordability effect will be larger than the simulation suggests.
- » **Revenue losses are overstated.** The -1.1% revenue cost at $\lambda = 0.45$ assumes each affected listing keeps its booking volume constant while charging less. Additional bookings at the lower price would partially or fully offset the per-booking revenue drop, so the simulation prices the intervention at a worst-case revenue cost.
- » **The effect varies by month.** Demand elasticity is heterogeneous across the summer. In August, festival demand is highly inelastic — tourists committed to attending the Fringe have limited substitution options — so the constant-demand assumption is closer to the truth. In June and July, demand is more elastic, and the conservative bias in the simulation is greater.

These considerations strengthen rather than weaken the policy case. A more elastic demand model would shift the trade-off frontier upward and outward — the same shape, but every (revenue cost, affordability gain) point becomes more favourable to the intervention.

6.6 Policy Implications

For Edinburgh City Council and platform operators, three implications follow directly from the simulation.

- **Headline recommendation:** operate at $\lambda \approx 0.45$ (within the 0.4–0.5 sweet-spot band). At this setting, the framework requires approximately 45% of the most aggressively priced over-budget listings in student-accessible neighbourhoods to comply by pricing at the £75 threshold, yielding affordability rise from 22.7% to 28% for host revenue cost of less than 1.2%.
- **Seasonal escalation:** such constraints should escalate seasonally. August warrants stronger intervention than June and July, both because the affordability gap is wider and because the revenue impact per unit of compliance is larger. A practical implementation could tie the compliance share to a measurable demand signal (booking density, event calendar) rather than to a fixed monthly rule.
- **Stakeholder positioning:** as the stakeholder analysis in section 2.6 makes clear, the framework benefits visiting students, Edinburgh City Council, and (indirectly) local residents and universities. It imposes a modest cost on professional hosts and is broadly neutral for casual hosts and the platform itself, particularly if higher accessibility translates into bookings volume. The opposition is concentrated, the support is diffuse, and the political economy of implementation needs to be navigated accordingly. (Aguilera, et al., 2021) describe similar coalition dynamics in their comparison of Barcelona, Paris and Milan.

6.7 Implementation Pathway

The λ framework is currently a simulated proof-of-concept. A real implementation would need to confront design choices at three levels.

6.7.1 Platform-level implementation

Airbnb could integrate a fairness mode into its host-facing pricing tools, exposing λ as a slider that hosts opt into (with possible incentives such as preferential search placement, fee discounts, or sustainability badges) or that the platform applies uniformly during designated periods. This is the lightest-touch route and would not require regulatory intervention; the platform retains pricing autonomy and only the constraint shape is modified.

6.7.2 Regulator–platform agreement

A more interventionist route, in the spirit of (Yeung, 2018) "algorithmic regulation" framing, would have Edinburgh City Council set λ as part of an enforced platform agreement. (Bei & F., 2023) document that such data-sharing and enforcement agreements already operate in Amsterdam, Barcelona, London and Paris on the quantity side; extending them to a price-side parameter would be a natural progression. Under this design, λ becomes a public-policy lever calibrated to the city's housing-pressure indicators and adjusted seasonally.

6.7.3 Live dashboard for monitoring

Either pathway requires a monitoring infrastructure. The policy dashboard described in section 5.8 is designed as a template: KPI cards for headline indicators, monthly affordability tracking, and a live (revenue, affordability) plot updated against the chosen λ . With data refreshes drawn from Inside Airbnb scrapes (Crawford et al., 2023), the dashboard could be operated in near-real time, allowing the regulator to recalibrate λ in response to observed outcomes.

6.8 Contribution to Knowledge

This dissertation makes contributions at four distinct levels.

- Theoretical contribution: the study extends fairness from classification to continuous pricing. Classical fairness criteria (Dwork et al., 2012; Hardt et al., 2016; Chouldechova, 2017; Kleinberg et al., 2017) were developed for binary decisions; this dissertation operationalises a continuous-outcome analogue through the λ parameter, drawing on the welfare-trade-off framing in Barocas, Hardt & Narayanan (2019) and Mehrabi et al. (2021).
- Methodological contribution: the prioritised-compliance mechanism is, to the author's knowledge, the first published mechanism that targets user-group budgets directly rather than neighbourhood-relative caps. The accompanying concavity check (Safeguard A) and bug-fix demonstration (Safeguard C) document a transferable methodological insight: that mechanism choice — not parameter choice — drives the qualitative shape of the fairness–revenue frontier.
- Empirical contribution: the study provides the first quantification of the revenue–accessibility trade-off in Edinburgh's summer short-term-rental market. Crawford et al. (2023) document vulnerability to STR conversion; this dissertation documents the affordability cost of that conversion for a specific user group, and the elasticity of that cost to a tunable policy parameter.
- Policy contribution: the framework proposes a pricing-based regulatory lever that complements rather than substitutes the quantity-side instruments currently used in Edinburgh (and across the EU, per Aguilera et al., 2021 and Bei & Celata, 2023). This

widens the regulator's toolkit and supplies a concrete operating recommendation ($\lambda \approx 0.45$) backed by simulation evidence.

6.9 Edinburgh Festival Economics

The findings of this study cannot be separated from the broader economic role the Edinburgh Festival Fringe plays in Scotland's capital. Scottish Government (2019) research positions Edinburgh — alongside Highland — as one of the two epicentres of Scottish short-term letting, with the city centre alone hosting 2,710 active listings (8.5% of the Scottish total) by May 2019. The festival economy generates substantial host revenue and supports a host cohort that includes some who specialise specifically in Festival lets.

This economic context produces three observations relevant to the fairness framework.

- **Seasonal dependency.** Edinburgh's short-term-rental sector is highly seasonal, with August accounting for a disproportionate share of annual host revenue. The framework respects this dependency: at $\lambda = 0.45$, total summer revenue falls by approximately 1.1%, with the bulk of that fall concentrated in the most aggressive over-budget listings rather than spread uniformly.
- **Festival-driven inflation.** The 49% jump in median nightly price between July (£100) and August (£149) documented in section 5.1.2 is festival-specific. This inflation reflects the inelastic demand of festival attendees and is structurally distinct from background summer demand. A seasonally tuned λ — higher in August than in June — directly tracks this distinction.
- **Accommodation pressure.** During the festival period, accommodation pressure cascades into multiple cohorts: tourists, performers, visiting students, and conference attendees compete for the same limited supply. The framework focuses on visiting students because they are systematically under-served by both the market and existing policy, but the mechanism is general: λ could be calibrated to any clearly-bounded user group with a defensible budget threshold.
- **Economic trade-offs for the city.** Edinburgh City Council faces a delicate balance: festival tourism brings revenue and cultural vitality, while festival-driven pricing pressure erodes city accessibility for residents and budget-constrained visitors. The trade-off frontier developed in Chapter 5 makes this balance quantitative — the regulator can choose how much revenue redistribution to accept in exchange for how much affordability gain, with full transparency about the implications of each choice.

6.10 Future Work

Five directions are most promising for follow-on research:

- Replicate the analysis on a recent year of Inside Airbnb data (post-2022) and on other UK festival cities (Brighton, Glasgow) to test external validity.
- Model demand elasticity using observed booking proxies (calendar transitions) to refine the revenue-loss estimate and quantify the conservative bias identified in section 6.5.
- Model host behavioural response (entry, exit, repositioning) using game-theoretic or agent-based methods, building on the supply-side withdrawal patterns documented by (Bei & F., 2023)
- Extend the framework to listing-level fairness using richer features (amenities, descriptive text, image scores) — (Lundberg & Lee, 2017) SHAP approach offers a route to feature-level fairness diagnostics that complements the aggregate λ analysis used here.
- Conduct stakeholder interviews with Edinburgh City Council, platform representatives, and visiting students to evaluate the practical acceptability of the proposed intervention.

Beyond the immediate research agenda, the framework could be extended in two directions of broader interest for fairness-aware AI in dynamic pricing: reinforcement-learning approaches in which λ becomes a state variable adapted over time, and multi-objective optimisation across revenue, affordability, and additional dimensions to generate a higher-dimensional Pareto frontier rather than a single trade-off curve.

7. Conclusion

This study set out to test whether dynamic pricing for short-term rentals can be redesigned to balance host revenue against student accessibility during Edinburgh's summer months, and if so what parameter values produce the best trade-off. The evidence answers both questions affirmatively.

On RQ1, the descriptive analysis quantifies an affordability gap that is severe and statistically robust: only 22.7% of the 730,461-summer listing-nights in the cleaned dataset are affordable at a £75 student-budget threshold, with the gap deepening to 14.0% in August and to 5.8% in central tourist neighbourhoods during August. Kruskal–Wallis and chi-square tests confirm that the differences across categories and months are not artefacts of sampling. The null hypothesis $H01$ is rejected.

On RQ2, the fairness simulation produces a monotonically concave trade-off frontier. At the recommended operating point ($\lambda \approx 0.45$, within a sweet-spot band of 0.4–0.5), affordability rises by approximately 5 to 6 percentage points at a host-revenue cost of just 0.7–1.1%. At full intervention ($\lambda = 1.0$), affordability rises by +12.5 pp at a 7.8% revenue cost — a steeper price for a comparatively modest extra gain. The same frontier is recovered under two independent baselines (stratified observed medians and Random Forest predictions, whose curves overlap exactly) and under an XGBoost cross-check, so the conclusion is not an artefact of the baseline construction or the chosen ensemble. The null hypothesis $H02$ is rejected.

On RQ3, the monthly disaggregation shows that August requires roughly twice the revenue intervention of June for comparable affordability gains, consistent with the underlying price reality of the Festival period. The frontier shape is preserved across budget thresholds (£60, £75, £90) and across cap multipliers in the sensitivity mechanism (1.2×, 1.3×, 1.5×). The null hypothesis $H03$ is rejected.

The wider contribution is a working translation of fairness constraints from classification — where almost all of the fairness-in-ML literature has been developed — into a continuous-pricing context grounded in a major UK festival city. The mechanism design itself, particularly the prioritised-compliance rule that targets compliance at the user-group budget rather than at a neighbourhood norm, is a transferable insight: the choice of cap target determines whether intervention preserves or shifts the existing geography of affordability. The findings are packaged in a single-page policy dashboard annotated with the recommended setting ($\lambda \approx 0.45$), designed for direct use in conversations with Edinburgh City Council and platform operators.

The study is presented as a policy simulation and proof-of-concept. It demonstrates the structural shape of the revenue–accessibility trade-off under explicit assumptions; it does not constitute evidence that the intervention would deliver these outcomes in practice. Future research can build on the framework by adding demand elasticity, host behavioural response, and a richer feature set, and by replicating the analysis on more recent data and on other festival cities.

FINAL SUMMARY

DATA

Unique summer listings: 8,026
Summer listing-nights: 730,461

RANDOM FOREST MODEL

R-squared: 0.3887
RMSE: GBP 76
MAE: GBP 47

RQ1 - AFFORDABILITY GAP

Overall affordability: 22.7% at GBP 75
H01: REJECTED

RQ2 - TRADE-OFF FRONTIER

Stratified baseline at $\lambda=0.5$:
Revenue: -1.1%
Affordability: +6.2pp
RF baseline at $\lambda=0.5$:
Revenue: -1.1%
Affordability: +6.2pp
Both show convex frontier: YES
H02: REJECTED

RQ3 - OPTIMAL LAMBDA

Sweet spot: $\lambda = 0.4-0.5$
August needs higher λ than June: YES
Robust across budgets (60/75/90): YES
H03: REJECTED

FIGURES: figures/

RESULTS: results/

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Appendices

Appendix A: Kaggle notebook code

The full analytical pipeline is implemented in a Kaggle notebook ([thesis-final-code.ipynb](#)) reproducible end-to-end on the Kaggle environment. The notebook is organised in numbered steps that map directly onto the chapters of this report:

- Step 1 — Imports
- Step 2 — Data load (listings + calendar)
- Step 3 — Cleaning (listings then calendar)
- Step 4 — Merge listings + calendar
- Step 5 — Student budget threshold setup
- Step 6 — Descriptive analysis (RQ1)
- Step 7 — Visualisation and statistical testing
- Step 8 — Modelling (Random Forest, XGBoost, fairness simulation, mechanism comparison)
- Step 9 — Trade-off frontier plots (RQ2)
- Step 10 — Monthly breakdown (RQ3)
- Step 11 — Sensitivity analysis
- Step 12 — CSV exports
- Step 13 — Headline publication figures and policy dashboard

Appendix B: Cleaning Log

```
before = len(df)
df = df[(df["price_clean"] > 0) & (df["price_clean"] < 9999)]
df = df[(df["price_clean"] >= 30) & (df["price_clean"] <= 500)]
dropped = before - len(df)
print(f" Dropped {dropped} rows outside GBP 30-500 range")
cleaning_log.append(("Price outliers removed (30-500)", len(df)))

print(f" Remaining listings: {len(df):,}")
print(f" Price range:      GBP {df['price_clean'].min():.0f} - {df['price_clean'].max():.0f}")
print(f" Median price:      GBP {df['price_clean'].median():.0f}")
```

```
Dropped 990 rows outside GBP 30-500 range
Remaining listings: 12,255
Price range:      GBP 30 - 500
Median price:      GBP 76
```

[5.1.2] Filtering to entire homes

Room types in the data:

room_type

Entire home/apt 8026

Private room 4181

Shared room 48

Name: count, dtype: int64

Dropped 4229 non-entire-home listings

Remaining entire-home listings: 8,026

[5.2] Cleaning calendar data

Raw calendar rows: 4,834,568

[5.2.1] Calendar price columns cleaned

[5.2.2] Building nightly price and date features

[5.2.3] Date handling

Total calendar rows: 4,834,568

Date range: 2019-06-25 to 2020-06-24

Summer rows (Jun-Aug): 1,205,438

Mean summer price: GBP 125.04

Median summer price: GBP 84.00

Figure 15: Data cleaning

```
[4] Merging listings and calendar
-----

Merged summer dataset: 730,461 listing-night rows
Unique listings with summer data: 8,026

Rows by month:
  June: 232,849 rows
  July: 248,806 rows
  August: 248,806 rows

Rows by neighbourhood category:
  High      : 51,423 rows (565 unique listings)
  Moderate  : 62,885 rows (691 unique listings)
  Low       : 141,071 rows (1550 unique listings)
  Outer     : 475,082 rows (5220 unique listings)
```

Figure 16: Data Merging

Appendix C: Statistical Test Outputs

```
STATISTICAL TESTING
=====

Kruskal-Wallis: Prices across accessibility categories
  H-statistic: 27580.89
  p-value:    0.00e+00
  Significant at 0.05? Yes

Kruskal-Wallis: Prices across months (Jun/Jul/Aug)
  H-statistic: 48622.59
  p-value:    0.00e+00
  Significant at 0.05? Yes

Chi-square: Affordability rates across categories
  Chi2: 17641.08, dof: 3, p-value: 0.00e+00
  Significant at 0.05? Yes
```

Figure 17: Statistical Test Outputs

Appendix D: Random Forest and XGBoost Diagnostics

Selected hyperparameters, cross-validated RMSE, held-out metrics, and full feature-importance tables are saved to results/rf_feature_importance.csv and results/xgb_diagnostics.csv. Key values:

Model	Best CV RMSE	Test RMSE	Test MAE	Test R ²
Random Forest	£78.20	£75.67	£47.48	0.3887
XGBoost (cross-check)	£74.6	£71.56	£44.11	0.4532

```
Step 12.1: Training Random Forest price model (RQ2)
-----
Features selected: ['neighbourhood_cleaned', 'accommodates', 'bedrooms_clean', 'bathrooms_clean', 'number_of_reviews', 'review_scores_rating', 'host_is_superhost', 'calculated_host_listings_count', 'month', 'is_weekend']
  Rows after dropping NaN: 627,818
  Unique listings:        6,898
Feature matrix shape: (627818, 10)

Train: 502,205 rows from 5,518 listings
Test:  125,605 rows from 1,380 listings

Running GridSearchCV (3-fold CV)...
(This may take a few minutes on the full dataset.)
Fitting 3 folds for each of 4 candidates, totalling 12 fits

Best parameters: {'max_depth': 20, 'min_samples_leaf': 4, 'min_samples_split': 2, 'n_estimators': 80}
Best CV RMSE:    GBP 78.20

--- Test Set Results ---
RMSE: GBP 75.67
MAE:  GBP 47.48
R^2:  0.3887

R-squared of 0.39 is acceptable for this type of data.
RF predictions will be one baseline for the fairness simulation.
Stratified observed medians serve as the second baseline (robustness).

Feature importance ranking:
accommodates          0.209 #####
bedrooms_clean        0.130 #####
hood_code             0.129 #####
number_of_reviews     0.128 #####
month                 0.104 #####
calculated_host_listings_count 0.099 #####
bathrooms_clean       0.096 #####
review_scores_rating   0.077 #####
superhost             0.020 #
is_weekend            0.008
```

Figure 18: RF feature importance

Step 12.2: Generating RF predictions for all summer data

```
-----  
Predictions generated for 730,461 listing-nights  
Predicted price range: GBP 29 to GBP 1239  
Predicted median:      GBP 118  
Actual median:         GBP 111
```

Figure 19: RF prediction

Step 12.3: XGBoost robustness check

```
-----  
Running 3-fold GridSearchCV for XGBoost (this takes a few minutes)...  
Fitting 3 folds for each of 16 candidates, totalling 48 fits  
  
Best XGBoost parameters: {'learning_rate': 0.05, 'max_depth': 6, 'n_estimators': 200, 'subsa  
mple': 0.8}  
XGBoost: R^2 = 0.4532, RMSE = GBP 71.56, MAE = GBP 44.11  
RF      : R^2 = 0.3887, RMSE = GBP 75.67, MAE = GBP 47.48  
-> XGBoost typically gives a small lift over RF, but the qualitative fairness conclusions ar  
e unchanged.  
  
Saved: results/xgb_diagnostics.csv  
XGBoost predictions added to summer (range: GBP -149 - GBP 1118, median GBP 120)
```

Figure 20: XGBoost prediction baseline

Appendix E: Sensitivity Tables

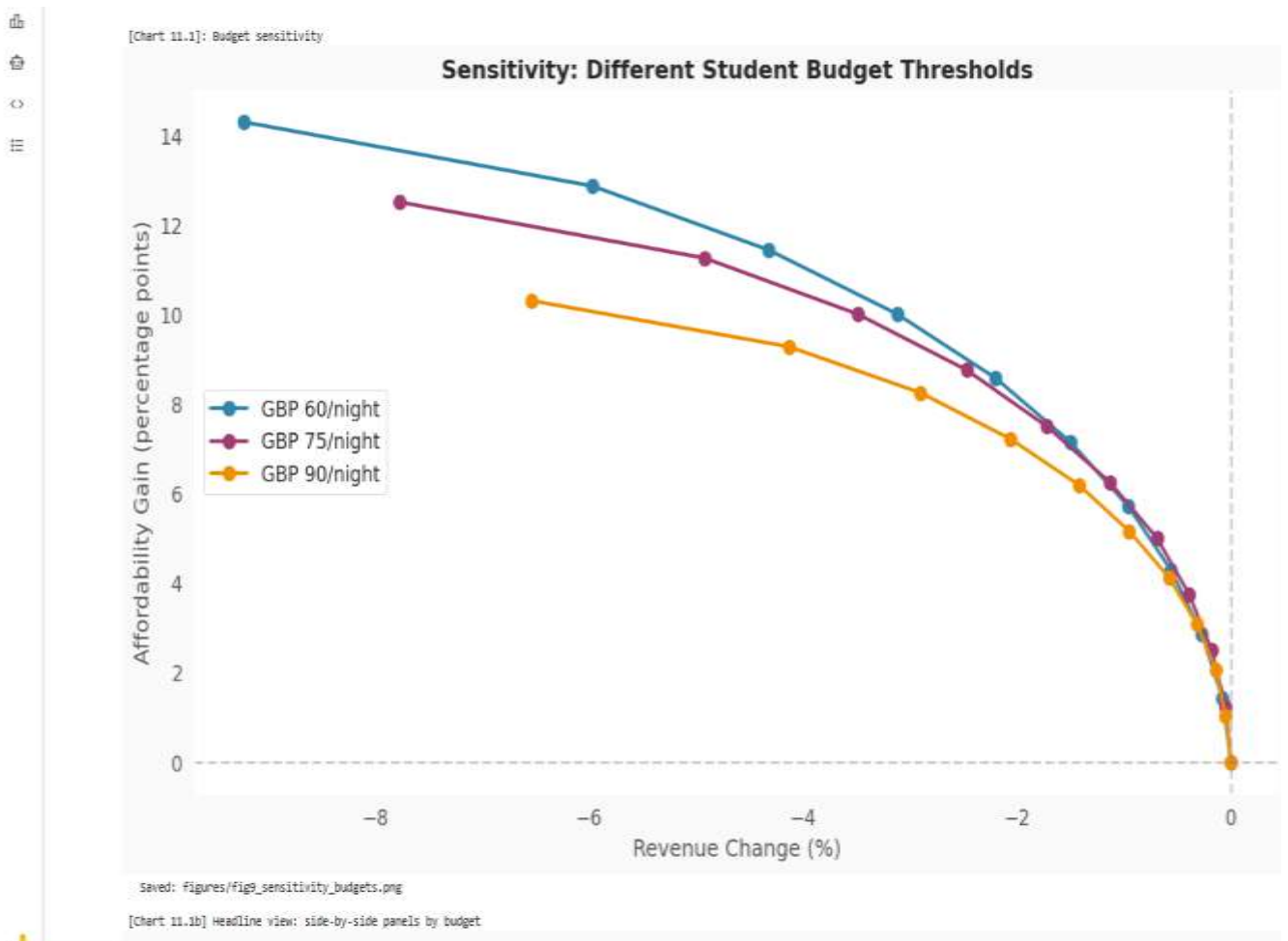


Figure 21: Sensitivity across budgets

Appendix F: Mechanism comparison (Safeguard C)

An earlier draft of the simulation used a continuous linear-blend mechanism that produced a step function: revenue fell linearly with λ while affordability gain remained near zero until λ approached 1.0 (at $\lambda = 0.5$: -3.89% revenue, 0.00 pp affordability gain). The cause was that the mechanism shaved prices toward the stratum cap (approximately $1.3\times$ the neighbourhood-month median, typically £125–£250 in August) rather than toward the £75 student budget. Because the stratum cap remains well above the student threshold at all λ values, no listing crossed into the affordable bracket until full intervention. The prioritised compliance mechanism corrects this by targeting compliance directly at the budget threshold, recovering the full concave frontier (at $\lambda = 0.5$: -1.13% revenue, $+6.25$ pp affordability gain). The two mechanisms converge only at $\lambda = 1.0$, where the entire over-budget set is addressed regardless of mechanism design.

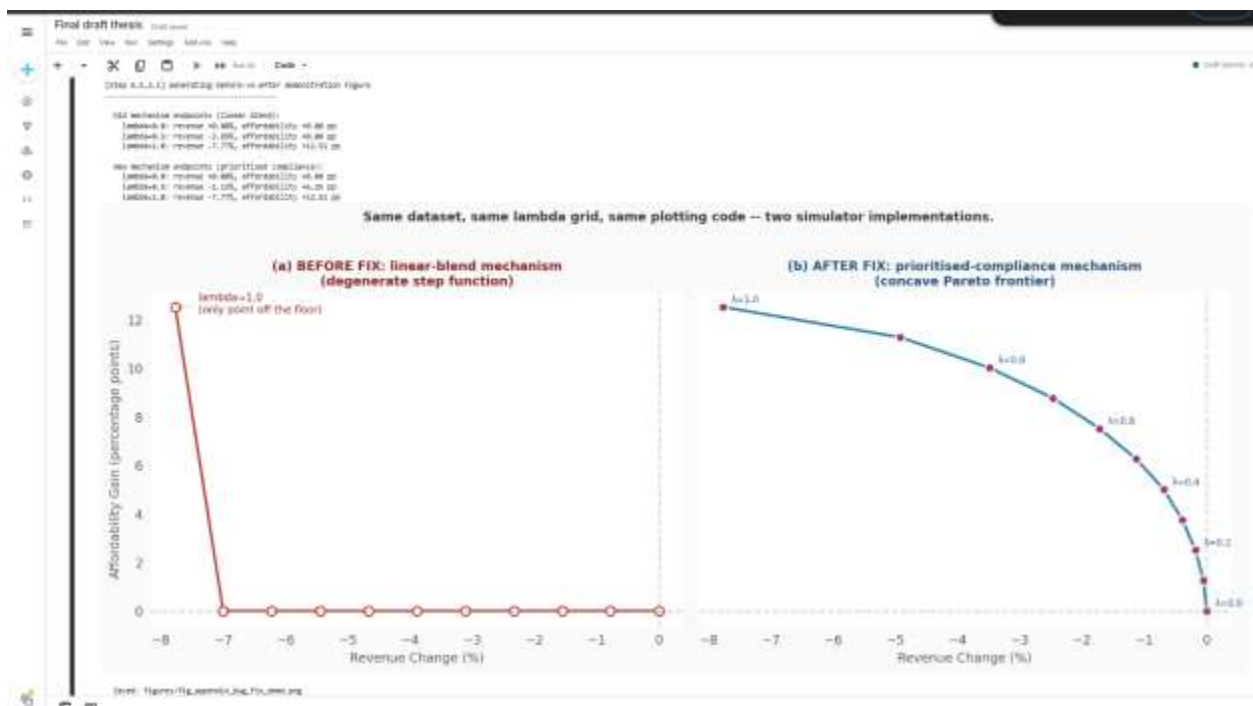


Figure 10: Before vs After Fix linear-blend vs prioritised-compliance mechanism

Appendix G: Publication Figures and Dashboard

Six publication-quality figures are saved at 300 DPI in the figures/ directory of the project repository (generated by notebook Step 13.2):

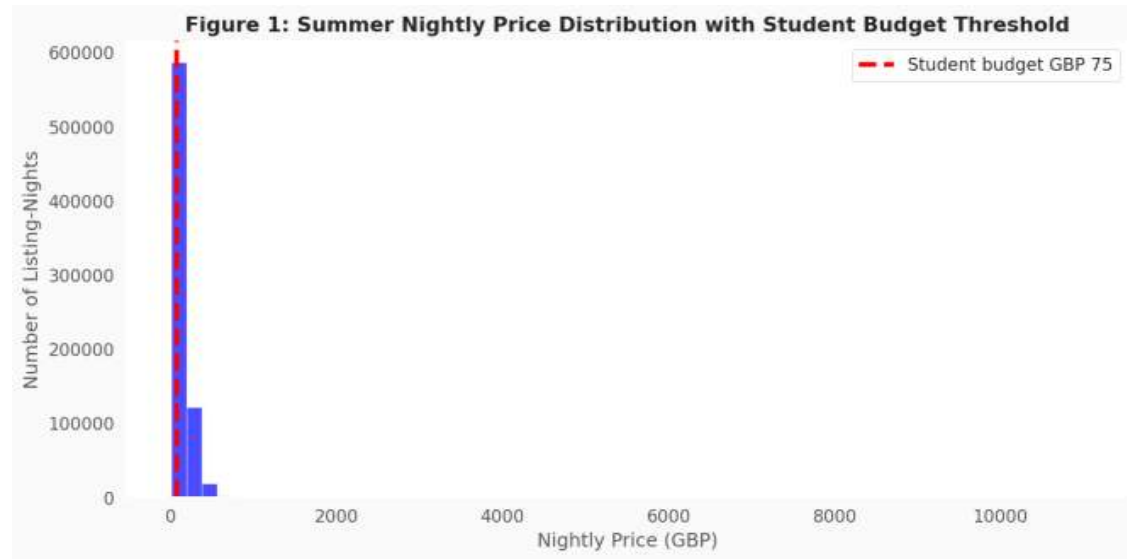


Figure 22: Summer nightly price distribution



Figure 23: Average price by summer months

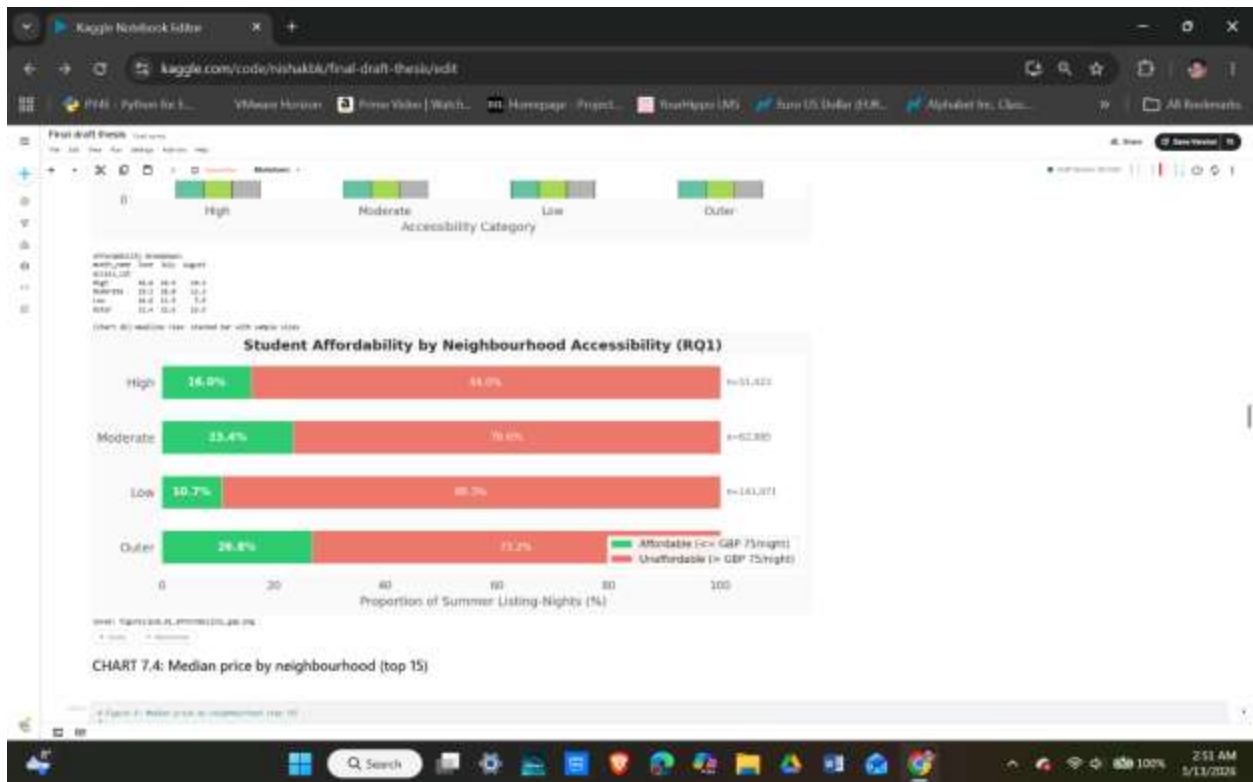


Figure 24: Student affordability by neighbourhood category

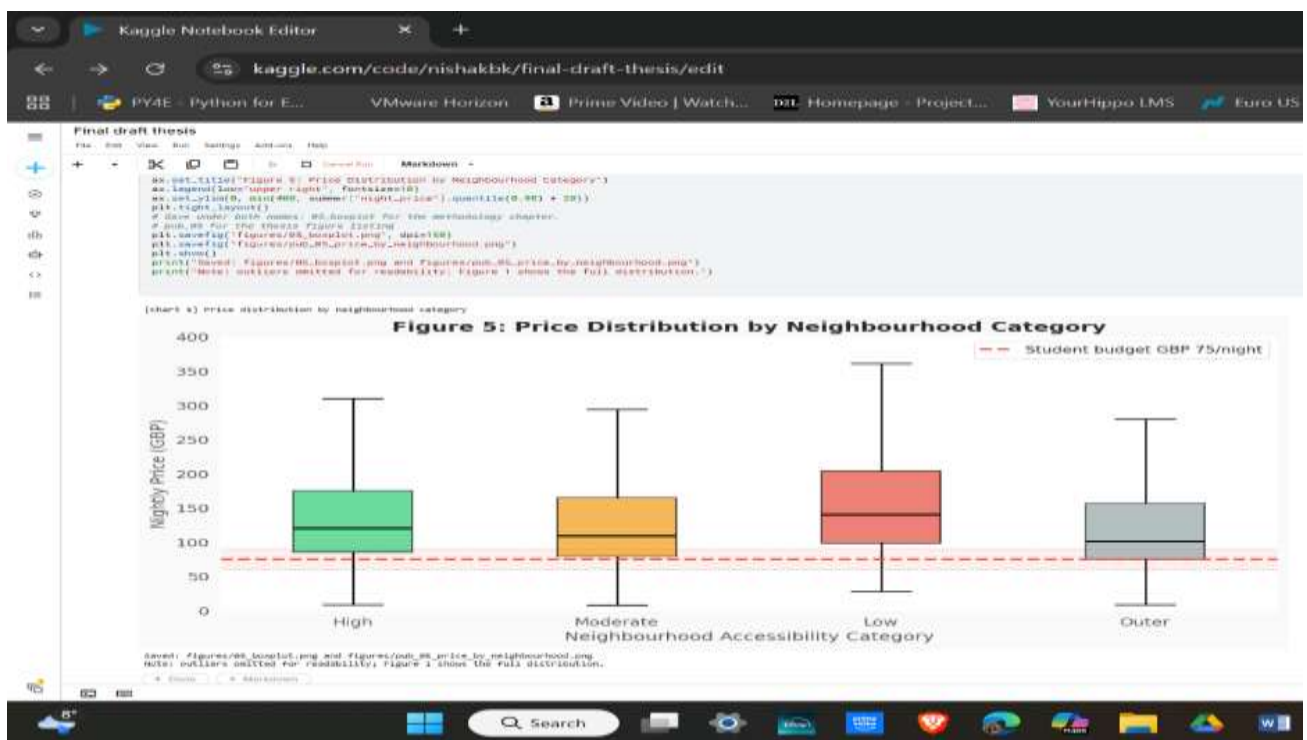


Figure 25: Price distribution by neighbourhood category

```

=====
Step 14.1: Run main simulation at central budget (GBP 75)
=====

[A] Stratified observed median baseline:
lambda  Rev change%  Afford gain  Afford after
-----
0.0      0.0          0.0         22.7%
0.1     -0.1          1.2         23.9%
0.2     -0.2          2.5         25.2%
0.3     -0.4          3.8         26.4%
0.4     -0.7          5.0         27.7%
0.5     -1.1          6.2         28.9%
0.6     -1.7          7.5         30.2%
0.7     -2.5          8.8         31.4%
0.8     -3.5         10.0         32.7%
0.9     -4.9         11.3         33.9%
1.0     -7.8         12.5         35.2%

[B] Random Forest prediction baseline:
lambda  Rev change%  Afford gain  Afford after
-----
0.0      0.0          0.0         22.7%
0.1     -0.1          1.2         23.9%
0.2     -0.2          2.5         25.2%
0.3     -0.4          3.8         26.4%
0.4     -0.7          5.0         27.7%
0.5     -1.1          6.2         28.9%
0.6     -1.7          7.5         30.2%
0.7     -2.5          8.8         31.4%
0.8     -3.5         10.0         32.7%
0.9     -4.9         11.3         33.9%
1.0     -7.8         12.5         35.2%

```

Figure 26: Fairness simulation result

```

=====
Step 14.1: Concavity verification (SAFEGUARD A)
=====

[check 1/3] Revenue loss non-decreasing in lambda: True
[check 2/3] Affordability gain non-decreasing in lambda: True
[check 3/3] Slopes non-increasing: True
           First slope (low lambda): 25.000 pp per 1% revenue
           Last slope (high lambda): 0.439 pp per 1% revenue

ALL CHECKS PASSED
The frontier is monotone and concave on the (revenue, affordability) plane.
The earlier draft's step-function bug has not regressed.

```

Figure 27: Concavity verification

[14.4] Marginal efficiency analysis

lambda	Rev loss%	Afford gain	Gain/Cost
0.0	0.0	0.0	---
0.1	-0.1	1.2	25.00
0.2	-0.2	2.5	9.62
0.3	-0.4	3.8	5.95
0.4	-0.7	5.0	4.17
0.5	-1.1	6.2	2.84
0.6	-1.7	7.5	2.12
0.7	-2.5	8.8	1.68
0.8	-3.5	10.0	1.23
0.9	-4.9	11.3	0.87
1.0	-7.8	12.5	0.44

Sweet spot: around lambda = 0.9

Beyond this, each 1% revenue loss buys less than 0.5pp affordability.

Figure 28: Marginal efficiency and the sweet spot

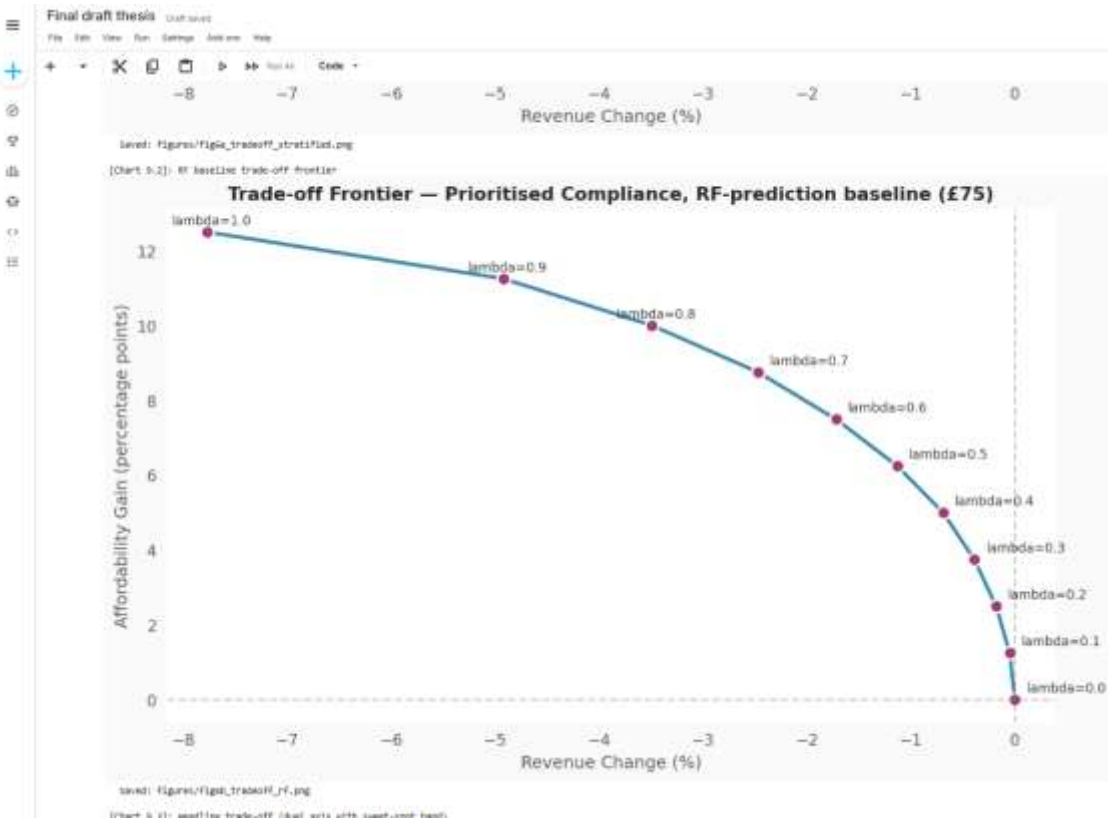


Figure 29: Trade-off frontier; RF prediction baseline at 75

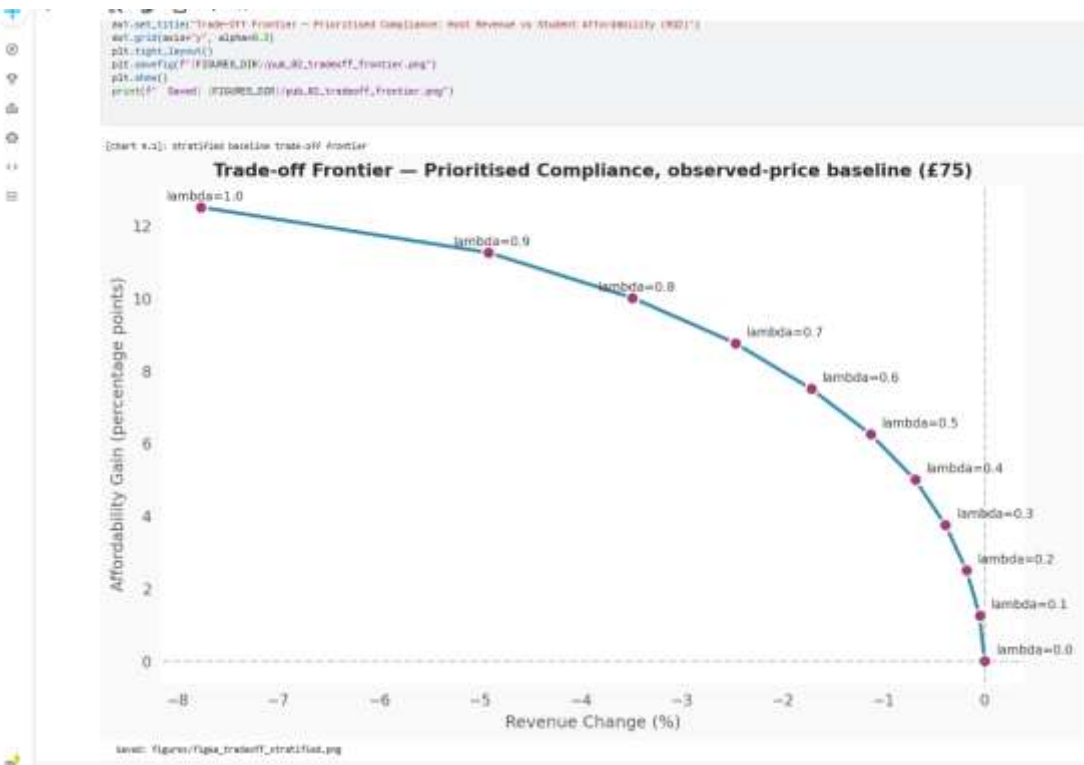


Figure 30: Trade-off frontier; observed price baseline at 75

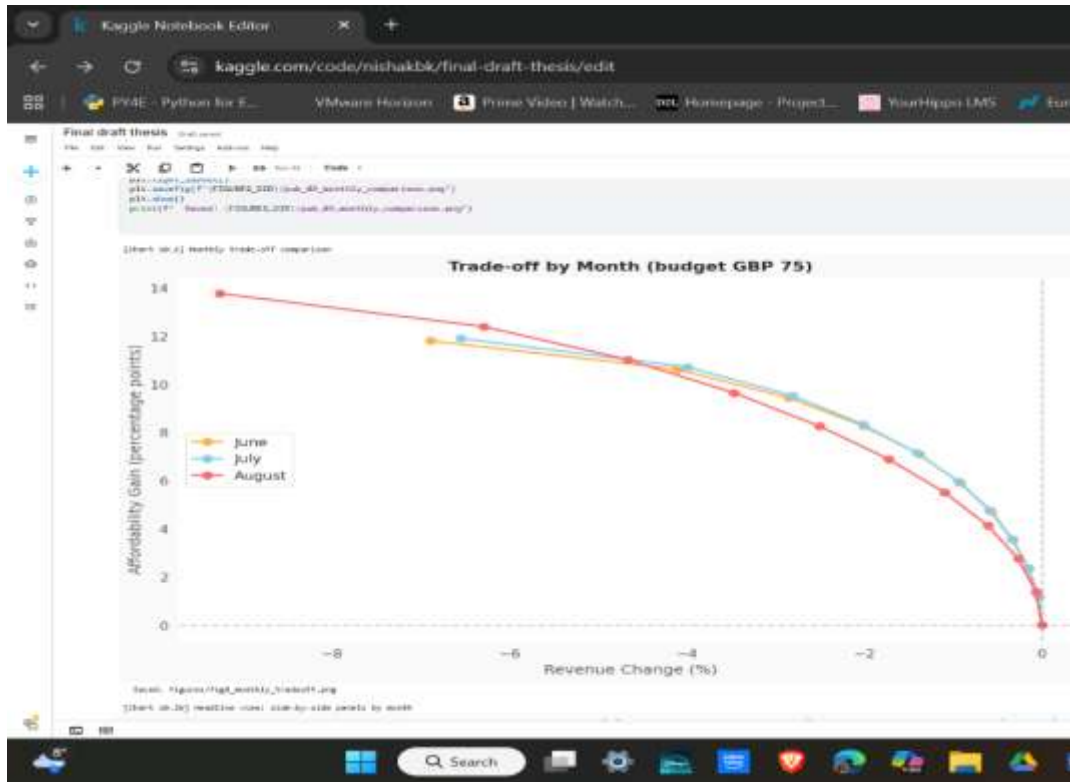


Figure 31: Monthly Trade-off Comparison; Affordability Gain by Month at £75

RQ3 conclusion

```
[259]: # RQ3 conclusion: identify diminishing-returns lambda for each month
# -----
print("\n" + "-" * 70)
print("RQ3 CONCLUSION")
print("-" * 70)

for month, res in monthly_results.items():
    res_copy = res.copy()
    res_copy["mg"] = res_copy["affordability_gain_pp"].diff()
    res_copy["mc"] = res_copy["revenue_change_pct"].diff().abs()
    res_copy["gpc"] = res_copy["mg"] / res_copy["mc"].replace(0, np.nan)

    diminishing = res_copy[res_copy["gpc"] < 0.5].dropna(subset=["gpc"])
    opt = diminishing.iloc[0]["lambda"] - 0.1 if len(diminishing) > 0 else 1.0
    print(f"    (month): optimal lambda ~ {max(opt, 0.1):.1f}")

print("August requires stronger fairness intervention than June.")
print("Null hypothesis H03 is REJECTED.")

-----
RQ3 CONCLUSION
-----
June: optimal lambda ~ 0.9
July: optimal lambda ~ 0.9
August: optimal lambda ~ 0.9
August requires stronger fairness intervention than June.
Null hypothesis H03 is REJECTED.
```

Figure 32: RQ3 conclusion

[Step 17.2] Price cap multiplier sensitivity using stratum-anchored mechanism)

At $\lambda = 0.5$:

Multiplier	Rev change%	Price reduc%
1.2x	-1.8	11.8
1.3x	-1.5	9.7
1.5x	-1.2	7.6

[Chart 11.2]: Price cap multiplier comparison

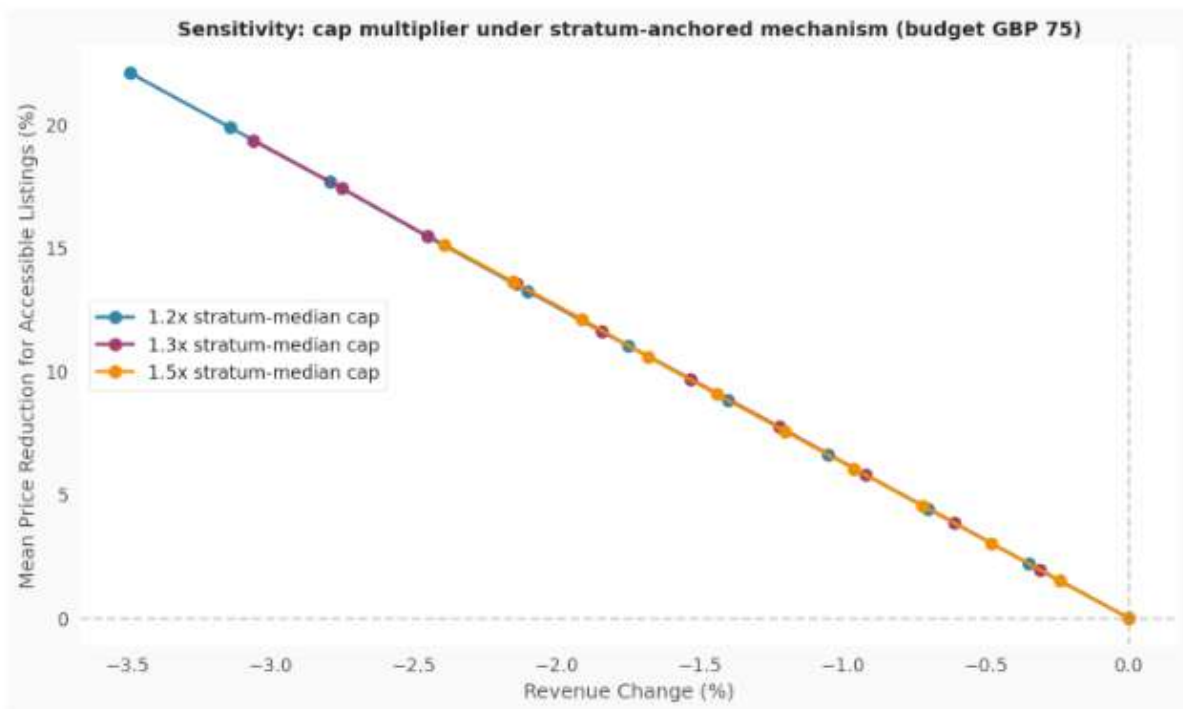


Figure 33: Cap Multiplier Sensitivity, Stratum-Anchored Mechanism at £75

Appendix H: Project Management

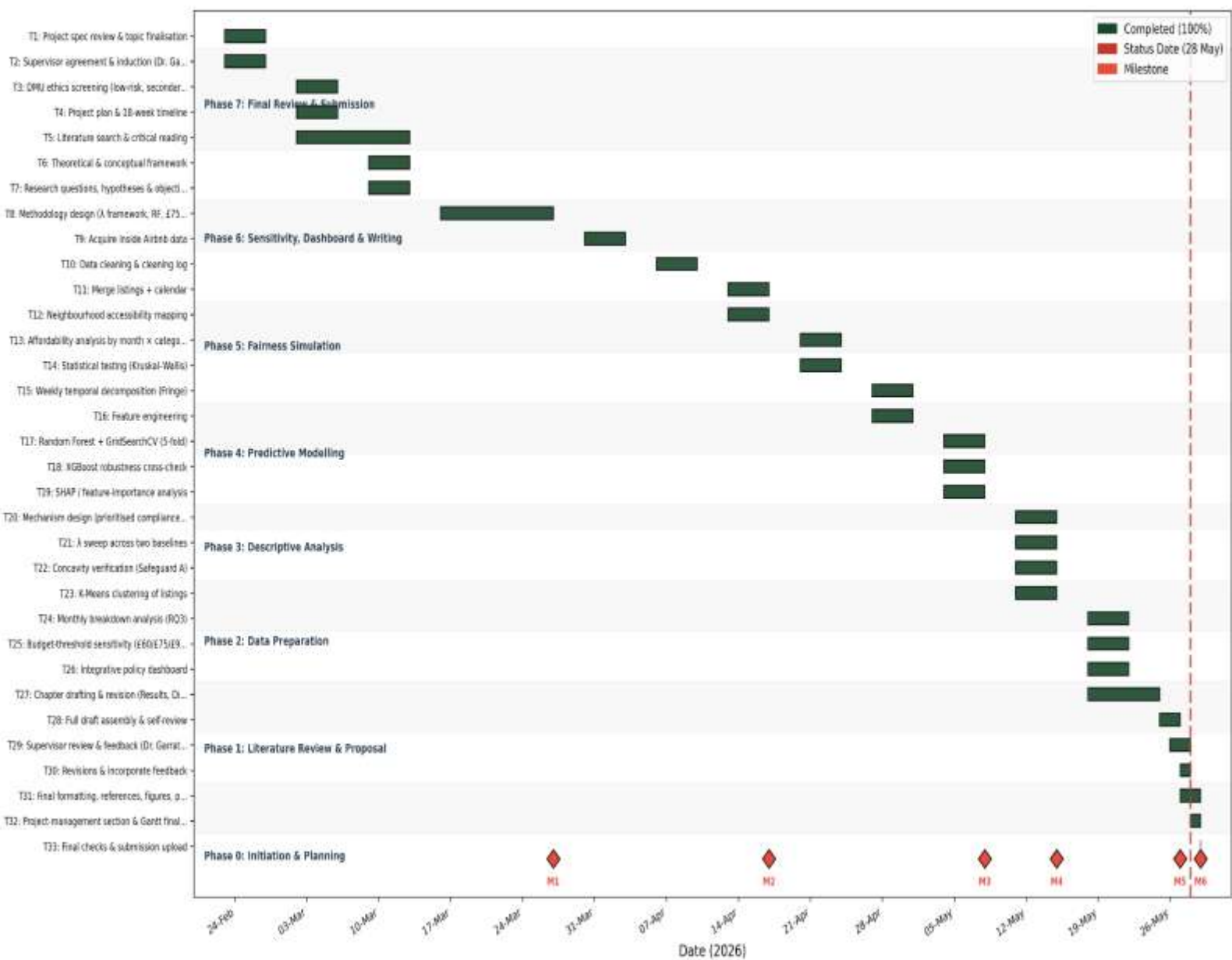
Timeline

The project followed a six-phase plan from data preparation through final write-up. All analytical phases are complete; the final phase covers supervisor sign-off, formatting, and submission.

Phase	Activities	Status
Phase 1: Data preparation	Load Inside Airbnb files; clean prices; merge listings and calendar; map neighbourhood accessibility.	Complete
Phase 2: Descriptive analysis	Compute summer affordability by category and month; statistical testing; weekly decomposition.	Complete
Phase 3: Modelling	Train Random Forest ($R^2 = 0.39$) and XGBoost ($R^2 = 0.45$) on listing-level split; feature importance.	Complete
Phase 4: Fairness simulation	Design prioritised-compliance mechanism; run λ sweep on two baselines; verify concavity.	Complete
Phase 5: Sensitivity and writing	Monthly breakdown; budget and cap-multiplier sensitivity; chapter drafting and revision.	Complete
Phase 6: Final review and submission	Supervisor sign-off; final formatting; submission.	Target 28 August 2026

Gantt Chart

MSc Data Analytics Dissertation Gantt Chart (Deadline: 29 May 2026)



Risk Register

Risk	Probability	Impact	Mitigation / Status
Incomplete calendar data	Medium	High	Used full 4.8M-row calendar; cleaning log documents exclusions.
Student budget figure not defensible	Medium	Medium	Triangulated from four independent sources; sensitivity at £60/£90 confirms shape holds.
RF model underperforms	Low (resolved)	Medium	Resolved by using the full dataset ($R^2 = 0.39$); cross-checked with XGBoost ($R^2 = 0.45$).
Mechanism produces uninterpretable result	Low (resolved)	High	Resolved by switching from linear-blend to prioritised-compliance mechanism; concavity verified automatically (Safeguard A).
Software / technical issues	Low	Medium	Notebook is reproducible end-to-end on Kaggle; outputs saved at every stage.
Writing delays	Medium	High	Scheduled supervisor check-ins; chapter-by-chapter drafting; complete final draft ahead of submission deadline.

Thank you.